AMITY INSTITUTE OF INFORMATION TECHNOLOGY

Course Code : CSE3048

"NOSQL DATABASES"

LABORATORY RECORD

Fall Semester - 2025

SUBMITTED TO:

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SUBMITTED BY:

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BCA

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EX.NO: 1

DATE: 12/11/2025

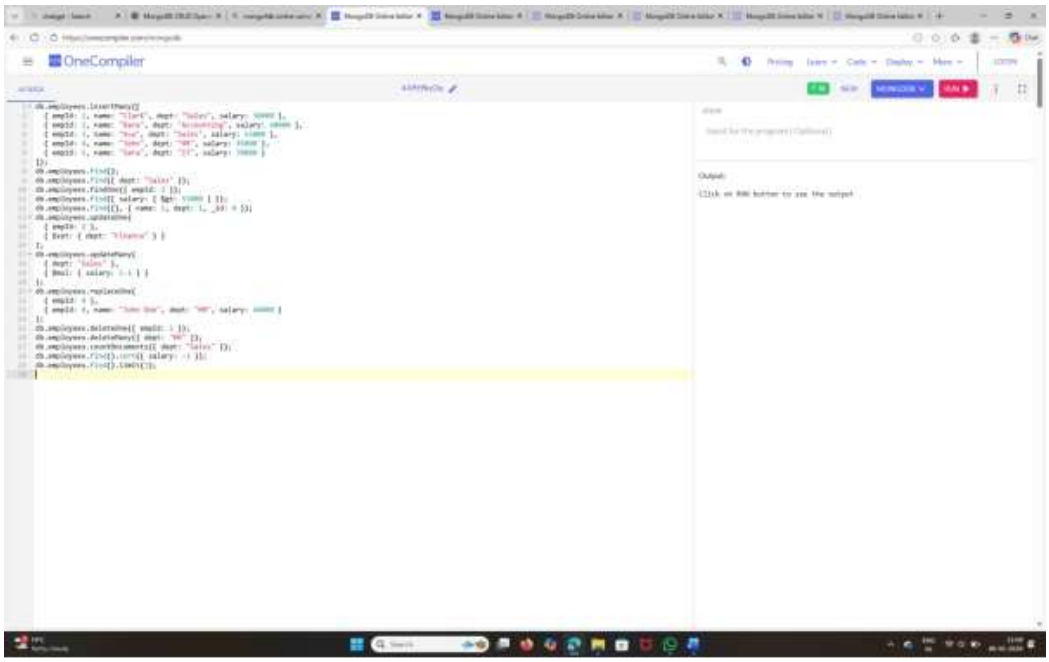
MongoDB CRUD Operations & Query Operators.

CODE:

```
db.employees.insertMany([
  { empid: 100, dept: "Accounting", salary: 8000 },
  { empid: 101, dept: "Marketing", salary: 9000 },
  { empid: 102, dept: "Sales", salary: 7000 },
  { empid: 103, dept: "IT", salary: 10000 }
]);
db.employees.find();
db.employees.find({ empid: 100 });
db.employees.find({ salary: { $gt: 9000 } });
db.employees.find({ dept: "IT" });
```

```
db.employees.find().sort({ salary: 1 });
db.employees.replaceOne(
  { empid: 100 },
  { empid: 100, dept: "HR", salary: 6000 }
);
db.employees.deleteOne({ empid: 101 });
db.employees.deleteMany({ salary: { $lt: 7000 } });
db.employees.find().limit(4);
```

INPUT



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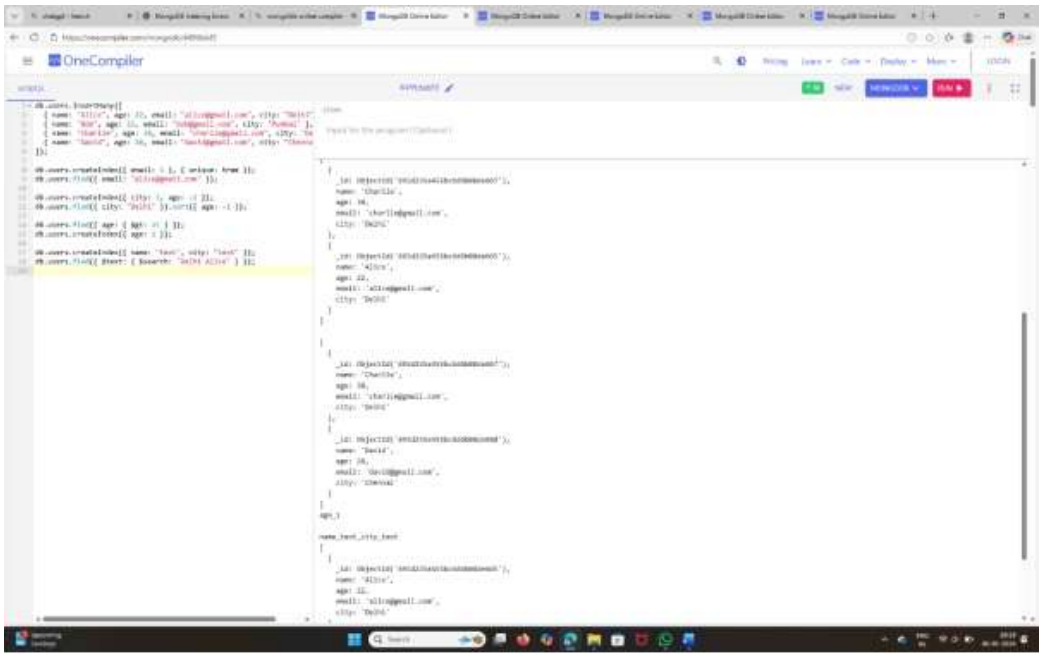

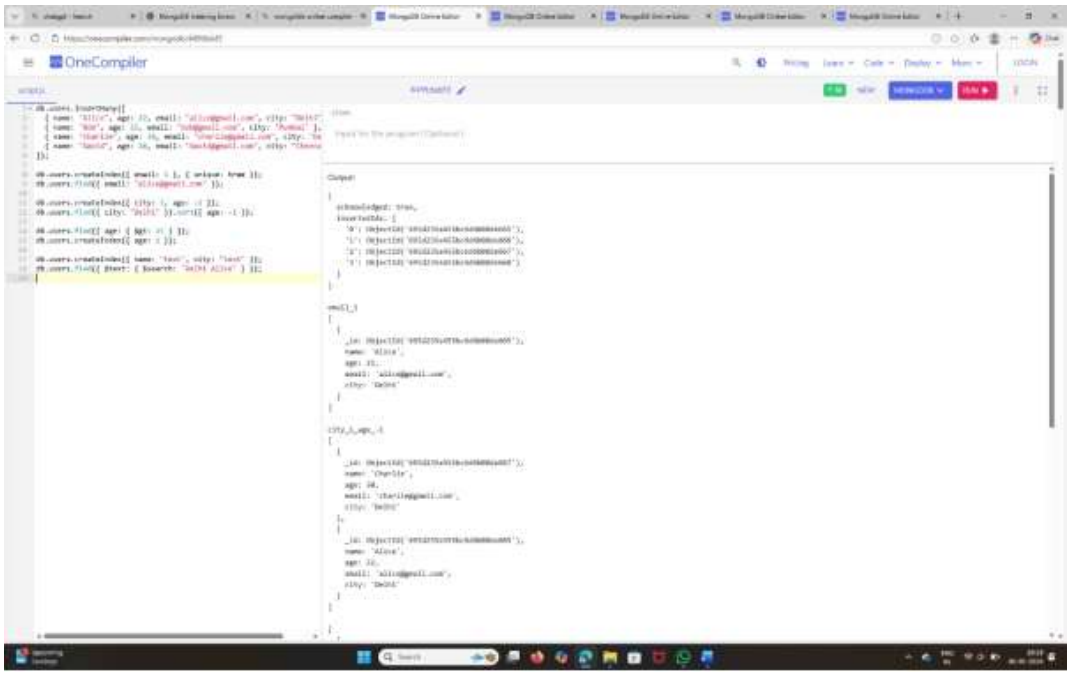

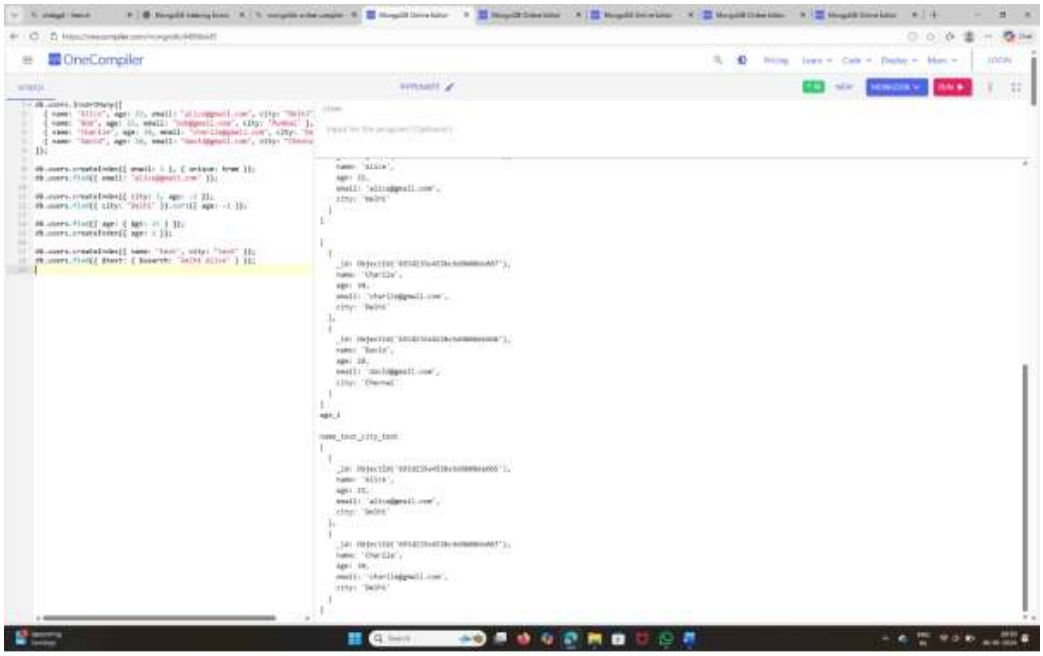
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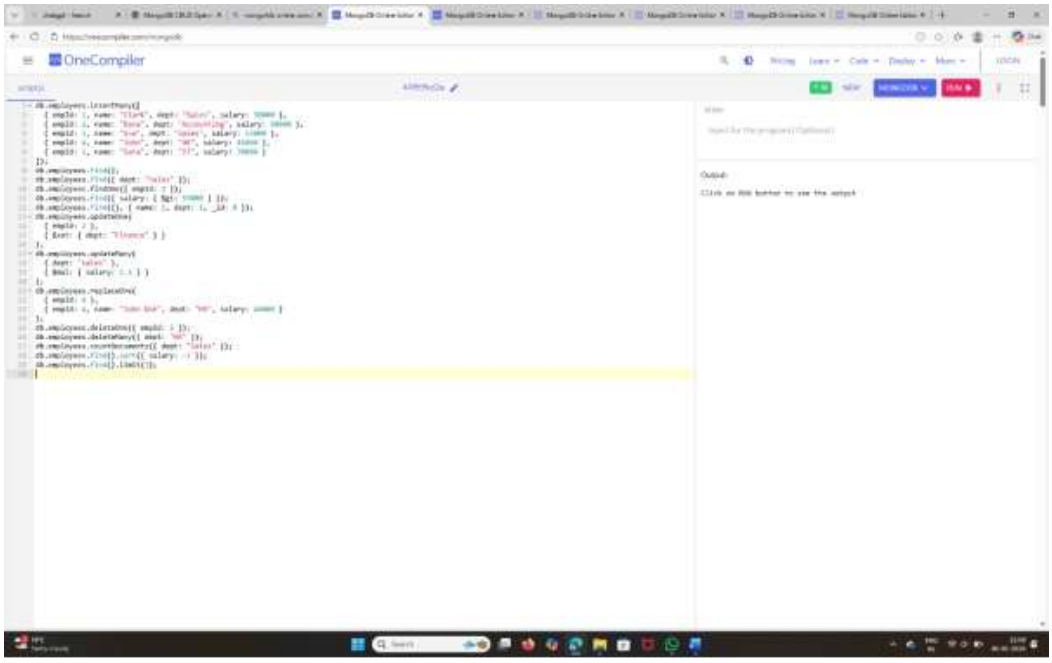
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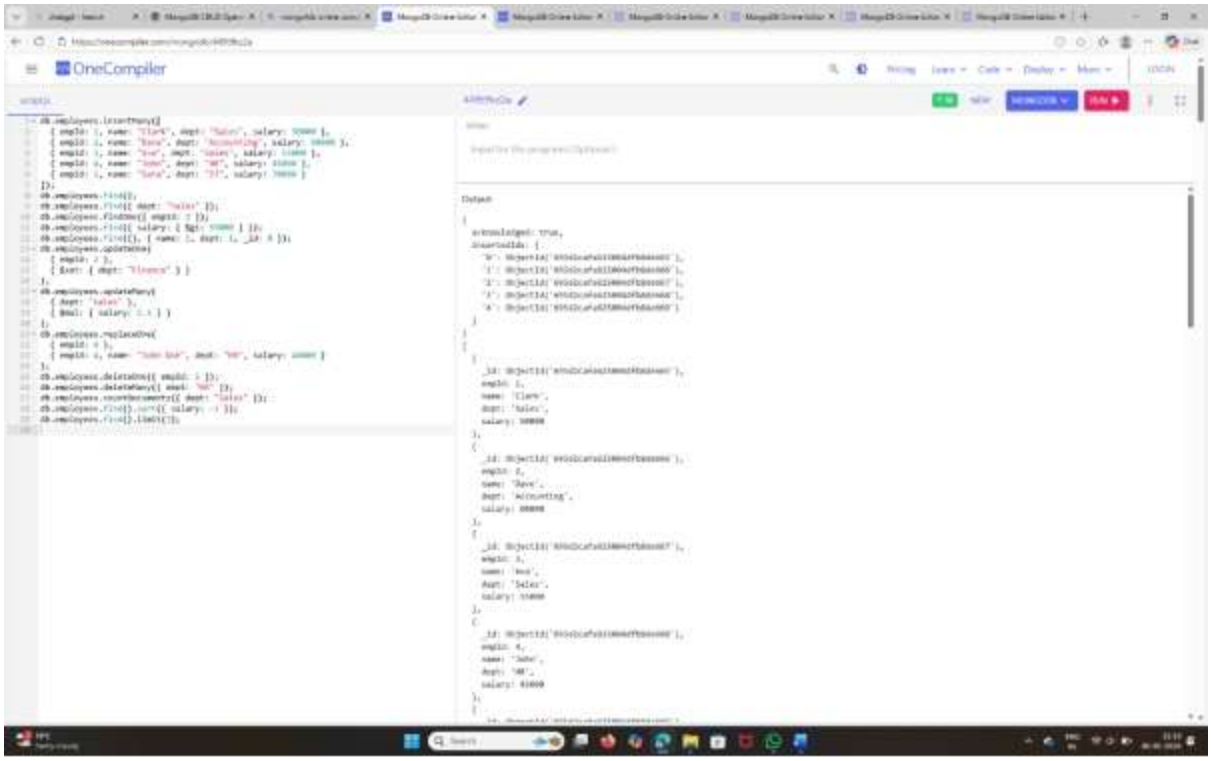
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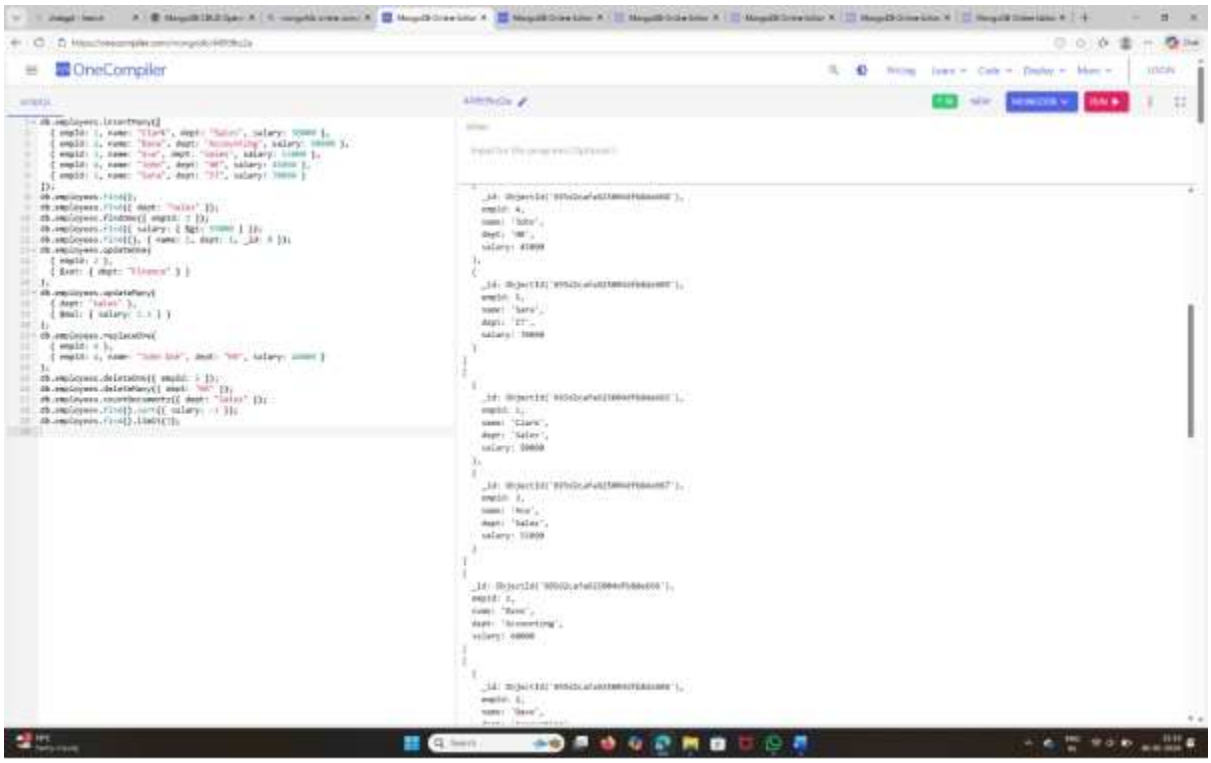
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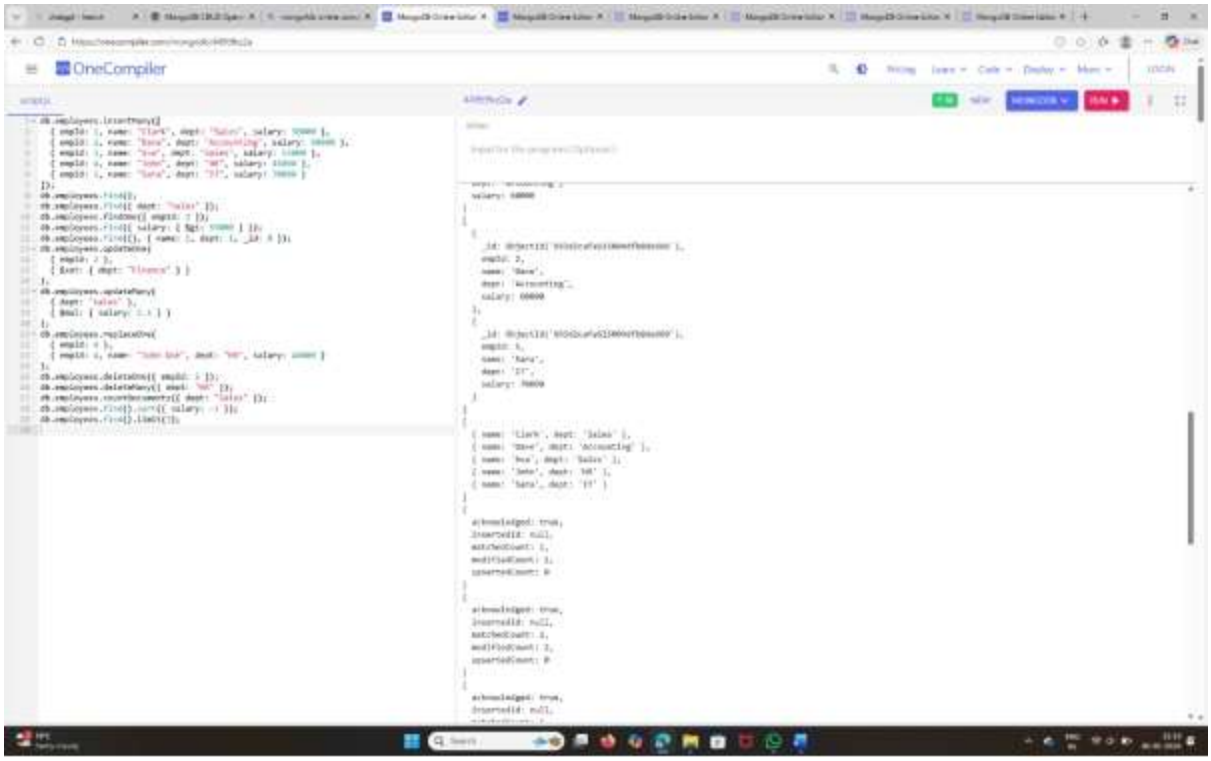
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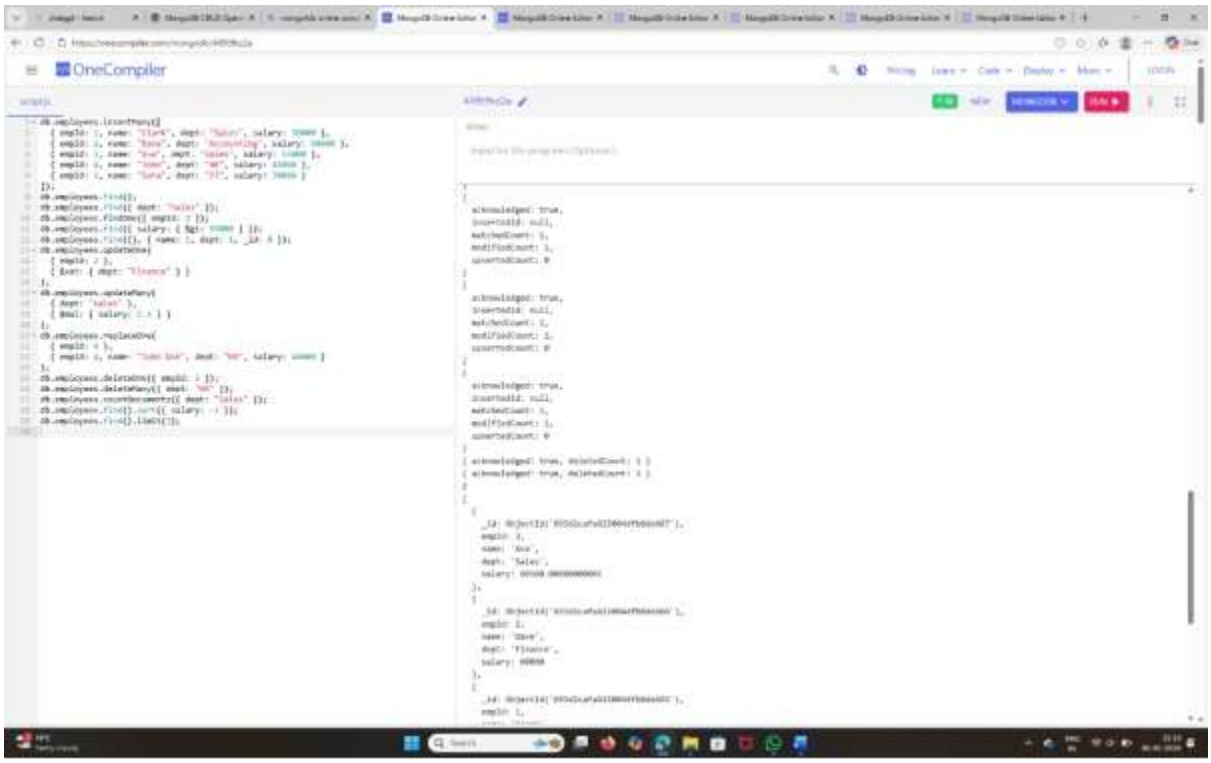
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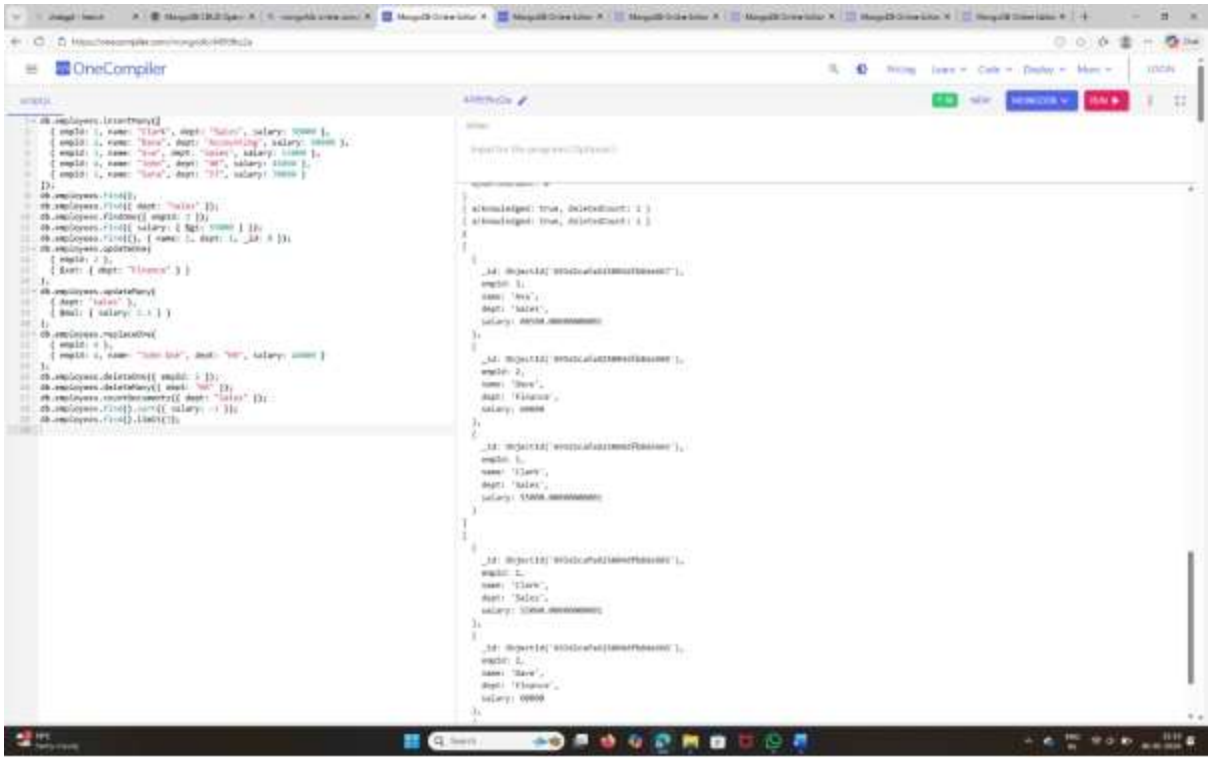
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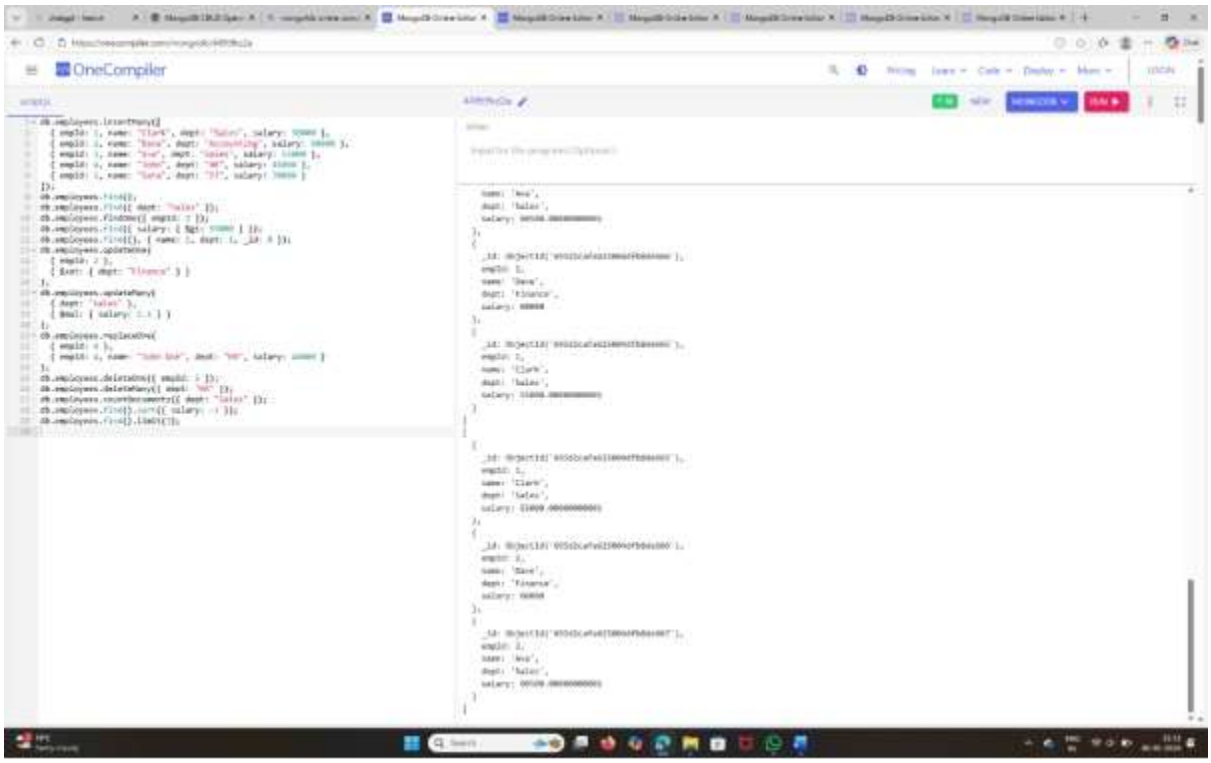

OneCompiler

EX.NO: 2

DATE: 17/11/2025

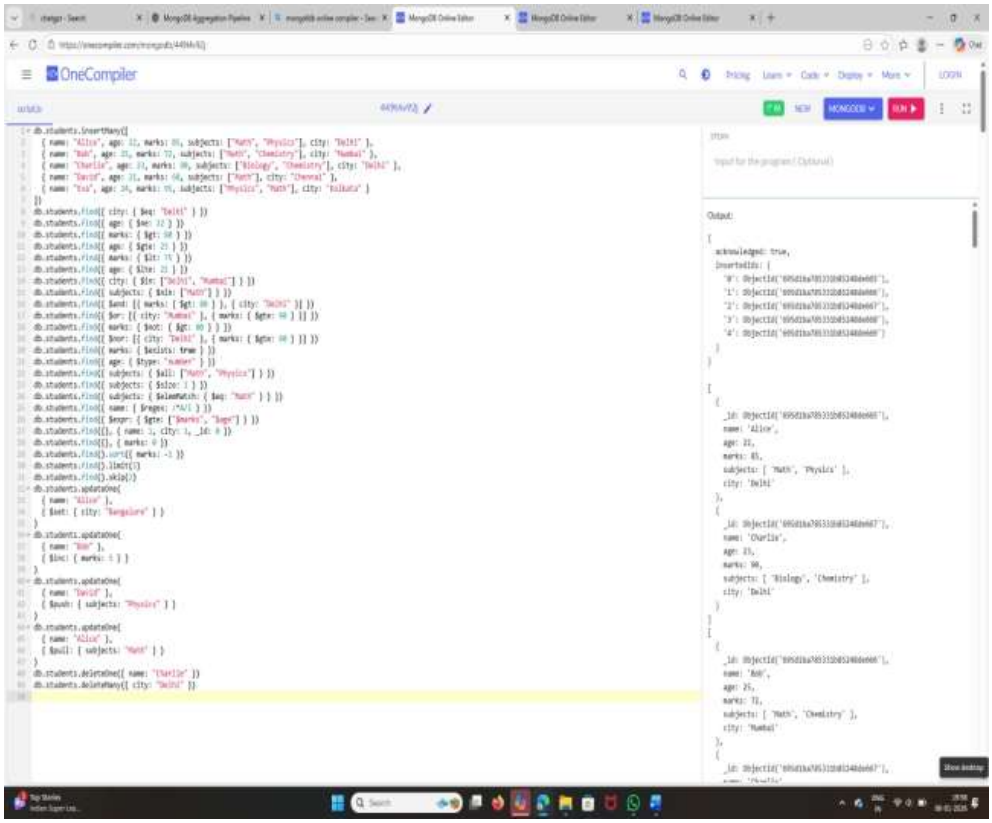
MongoDB CRUD Operations on Employees Collection.

CODE:

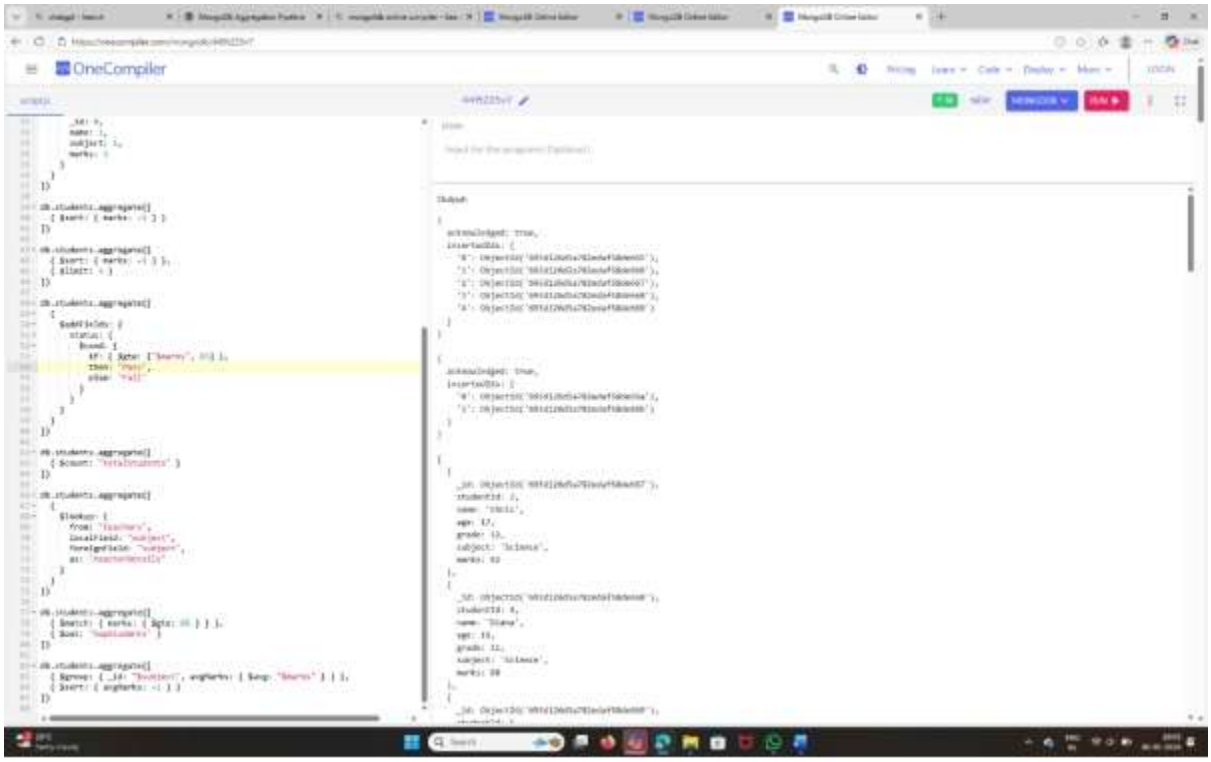
```
db.students.insertMany([
  { name: "Alice", age: 22, marks: 85, subjects: ["Math", "Physics"], city: "Delhi" },
  { name: "Bob", age: 25, marks: 72, subjects: ["Math", "Chemistry"], city: "Mumbai" },
  { name: "Charlie", age: 23, marks: 90, subjects: ["Biology", "Chemistry"], city: "Delhi" },
  { name: "David", age: 21, marks: 60, subjects: ["Math"], city: "Chennai" },
  { name: "Eve", age: 24, marks: 78, subjects: ["Physics", "Math"], city: "Kolkata" }
]);
db.students.find({ city: { $eq: "Delhi" } });
db.students.find({ marks: { $gt: 80 } });
db.students.find({ age: { $gte: 23 } });
db.students.find({ age: { $lte: 23 } });
db.students.find({ city: { $in: ["Delhi", "Mumbai"] } });
db.students.find({ subjects: { $all: ["Math"] } });
db.students.find({ city: "Delhi", marks: { $gte: 80 } });
db.students.find().sort({ marks: -1 });
db.students.find({ city: "Delhi" }).sort({ marks: 1 });
db.students.find({ city: "Mumbai" }).sort({ age: -1 });
db.students.find({ age: { $type: "number" } });
db.students.find({ subjects: { $size: 2 } });
db.students.find({ subjects: { $in: ["Physics"] } });
db.students.find({ subjects: { $nin: ["Math"] } });
db.students.find({ name: { $regex: "^A" } });
db.students.find({ name: { $regex: "e$" } });
```

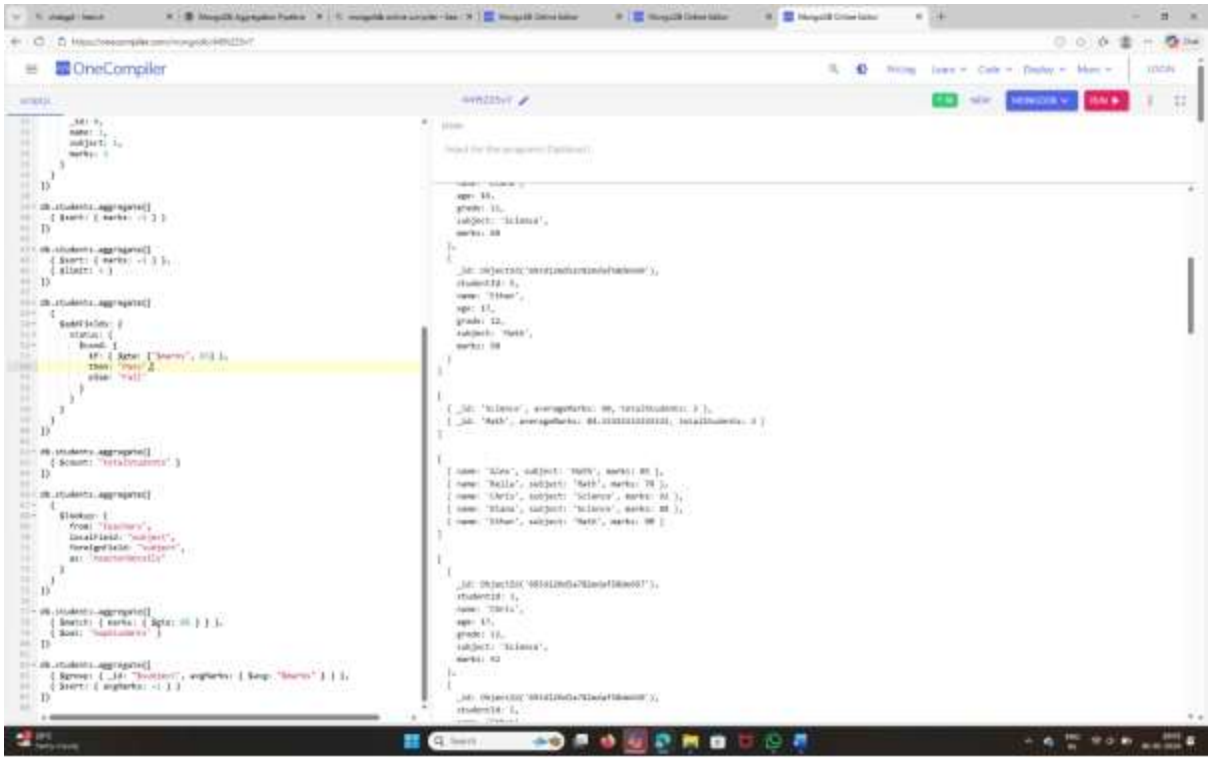
```
db.students.find({}, { name: 1, city: 1, _id: 0 });
db.students.find({ name: "Bob" });
db.students.find().count();
db.students.find().limit(3);
db.students.find().skip(2);
db.students.updateOne(
  { name: "Alice" },
  { $set: { city: "Bangalore" } }
);

db.students.updateOne(
  { name: "Bob" },
  { $inc: { marks: 5 } }
);
db.students.updateOne(
  { name: "David" },
  { $push: { subjects: "Physics" } }
);
db.students.updateOne(
  { name: "Alice" },
  { $pull: { subjects: "Math" } }
);
db.students.deleteOne({ name: "Charlie" });
db.students.deleteMany({ city: "Delhi" });
```

INPUT



OUTPUT

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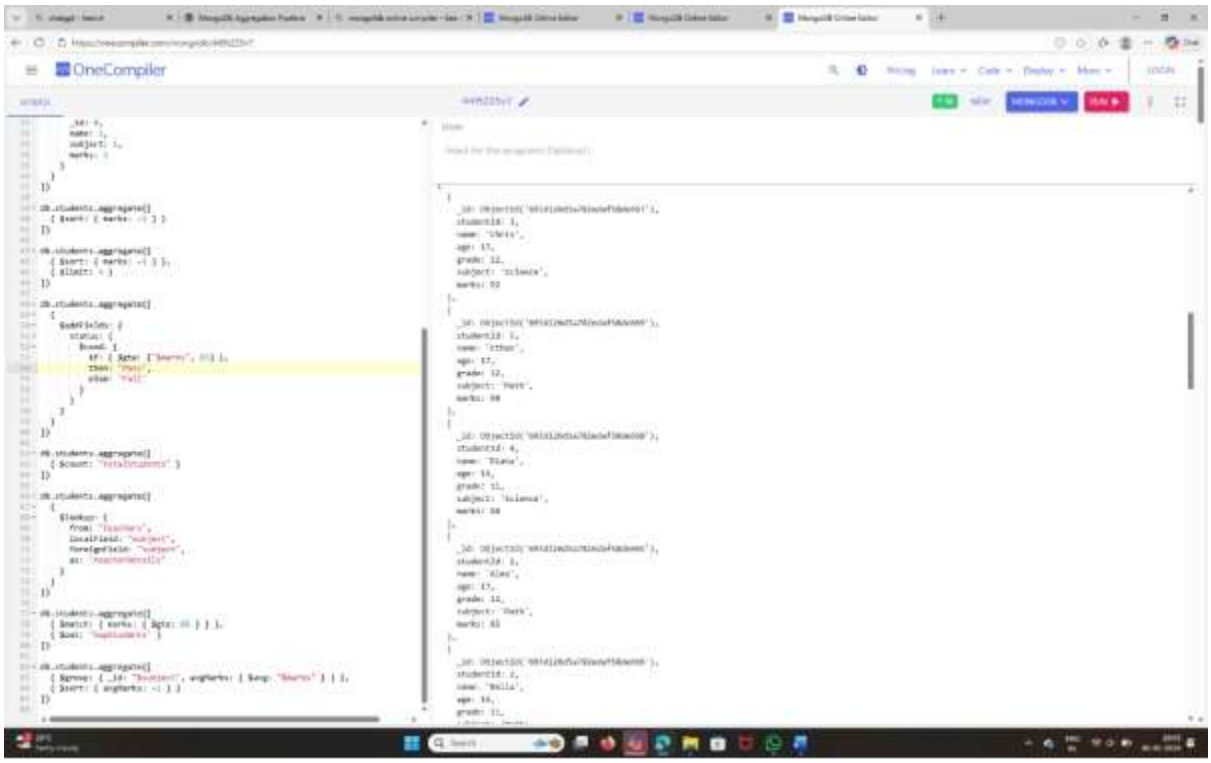
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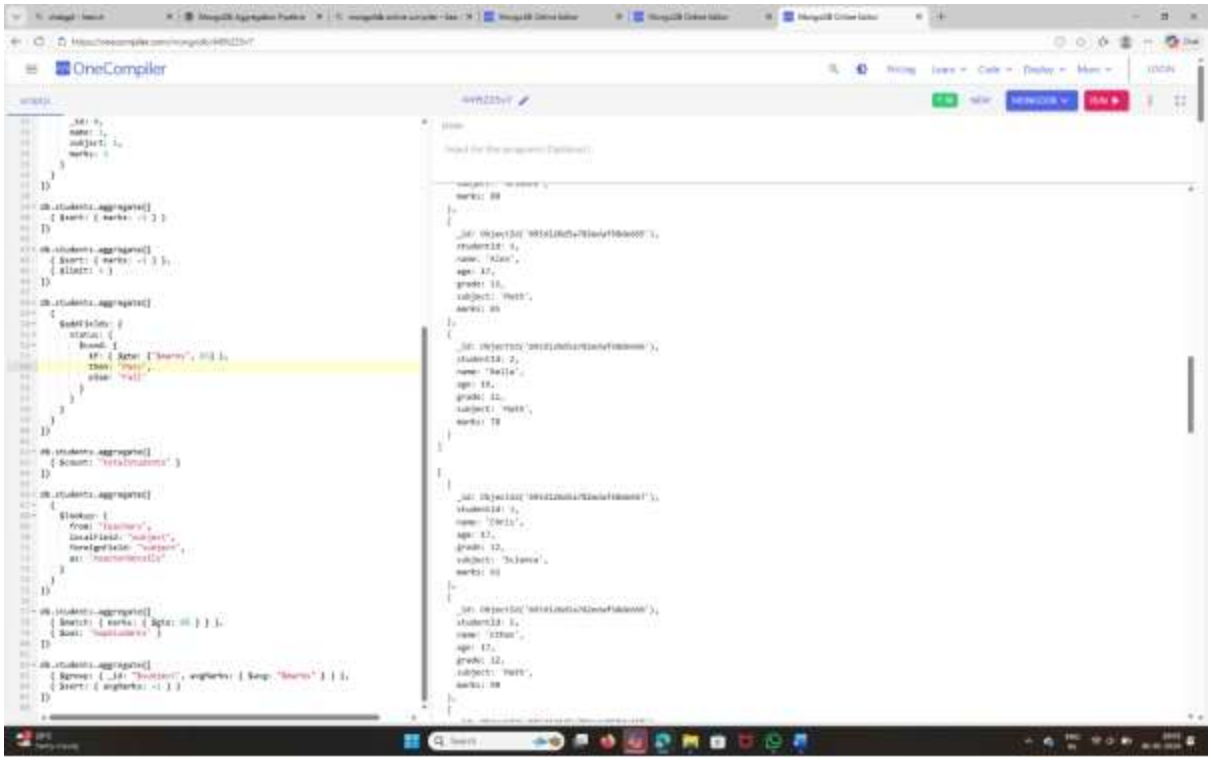
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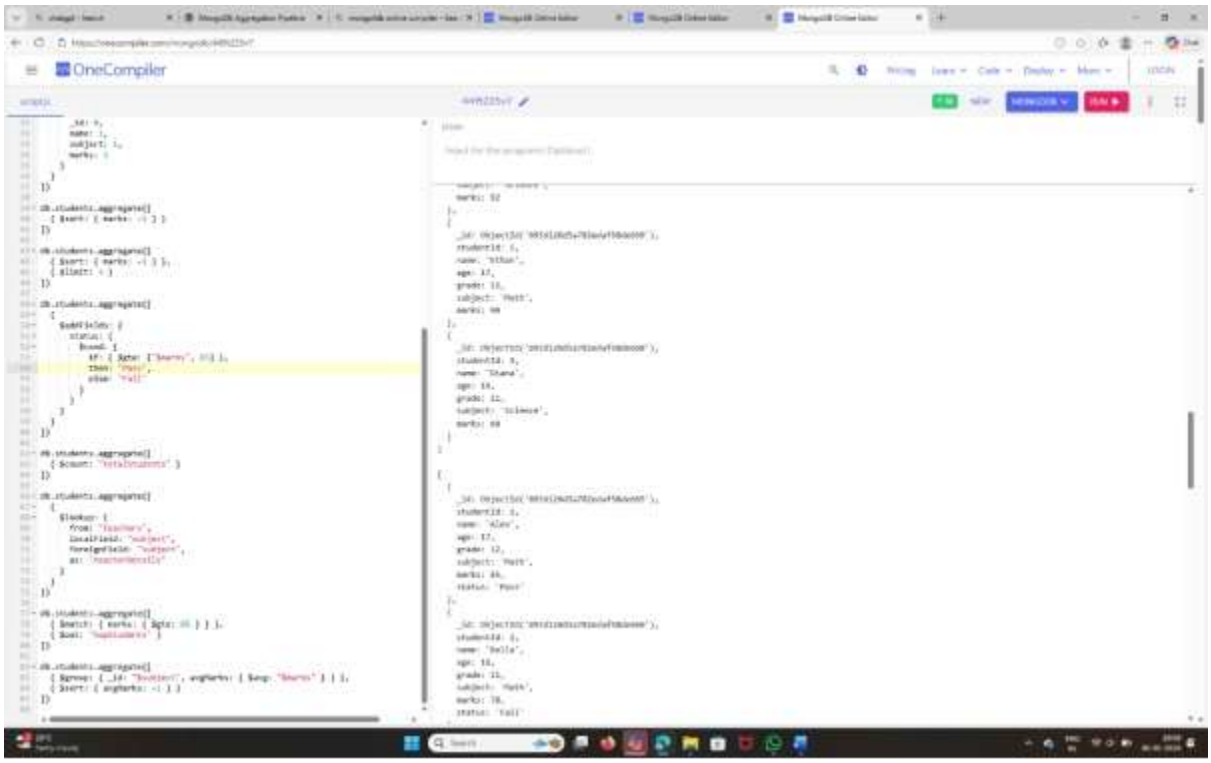
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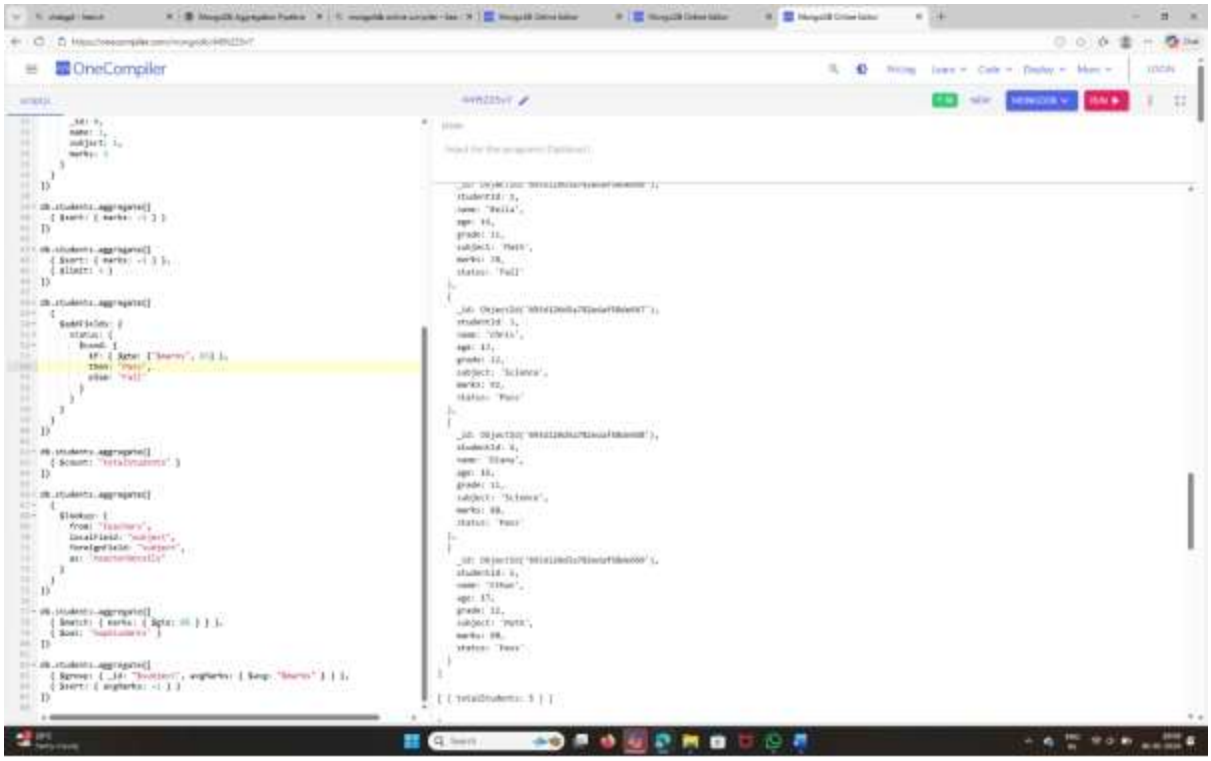
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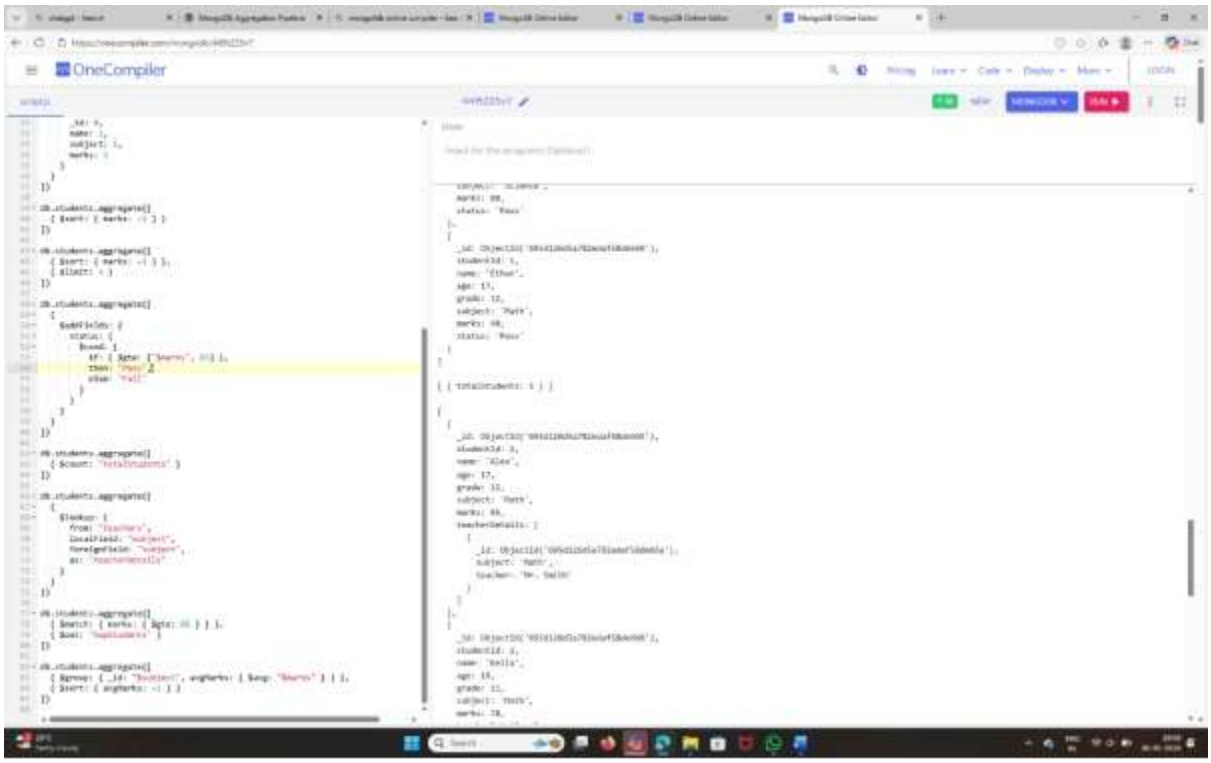
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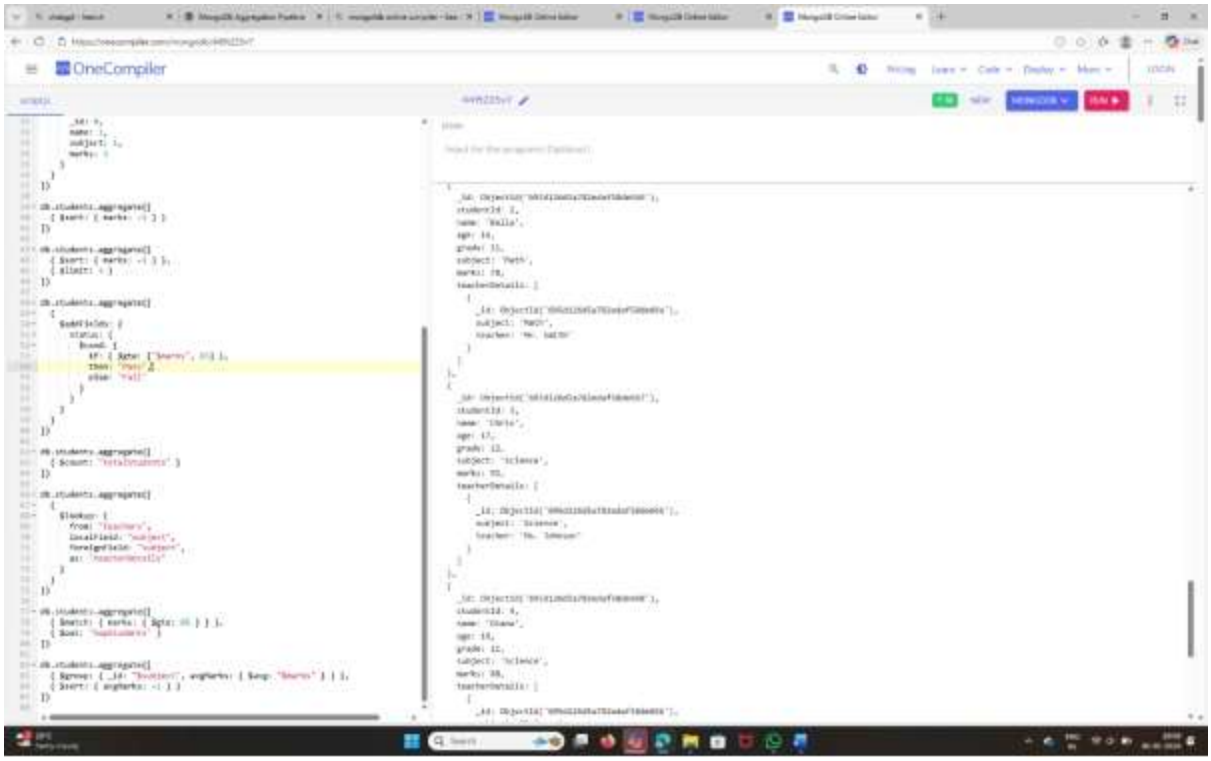
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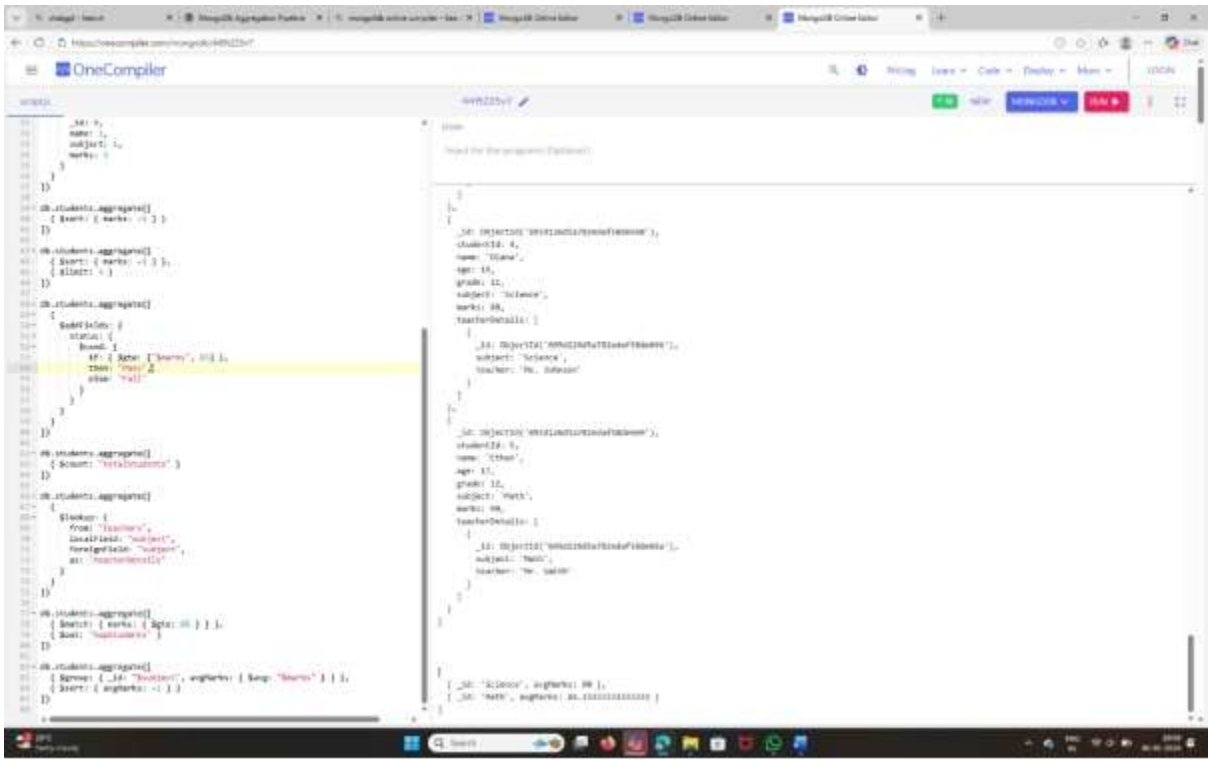

OneCompiler

EX.NO: 3

DATE: 25/11/2025

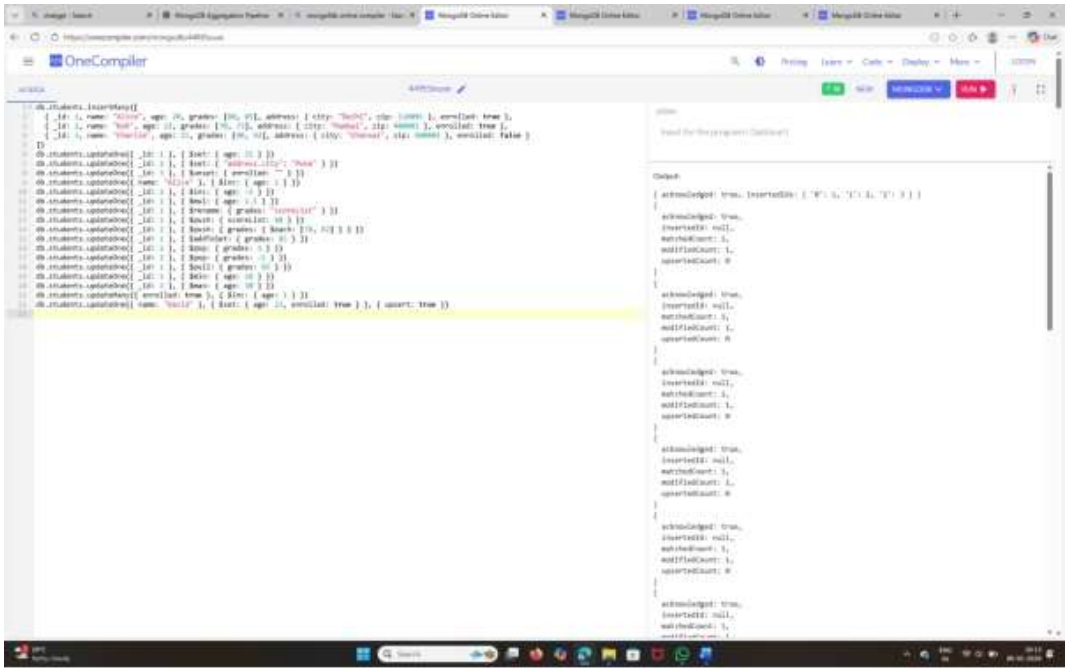
MongoDB UPDATE Operators on Students Collection.

CODE:

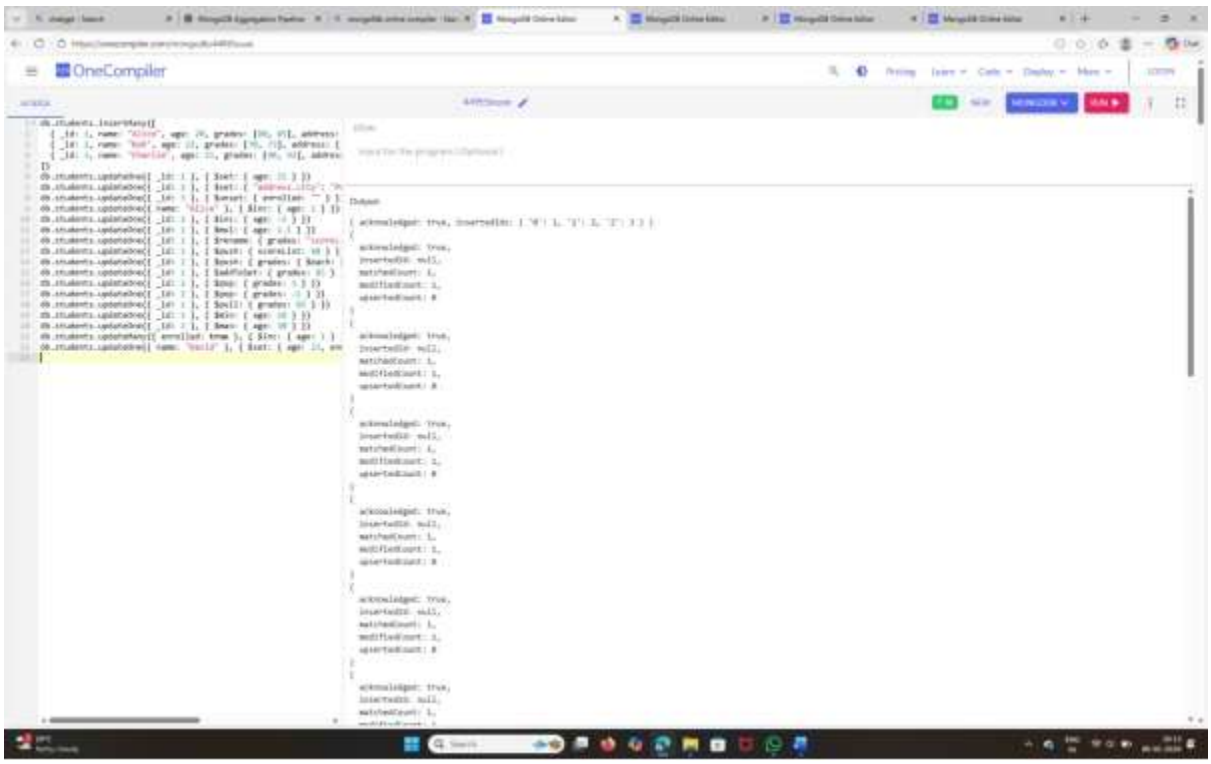
```
db.students.insertMany([
  { name: "Alice", age: 22, grades: [85, 90], address: { city: "Delhi", zip: 110001 }, enrolled: true },
  { name: "Bob", age: 25, grades: [70, 75], address: { city: "Mumbai", zip: 400001 }, enrolled: false },
  { name: "Charlie", age: 23, grades: [88, 92], address: { city: "Delhi", zip: 110002 }, enrolled: true }
]);
db.students.updateOne(
  { name: "Alice" },
  { $set: { age: 23 } }
);
db.students.updateOne(
  { name: "Bob" },
  { $inc: { age: 1 } }
);
db.students.updateMany(
  { address.city: "Delhi" },
  { $set: { enrolled: false } }
);
db.students.updateOne(
  { name: "Charlie" },
  { $push: { grades: 95 } }
);
db.students.updateOne(
  { name: "Alice" },
```

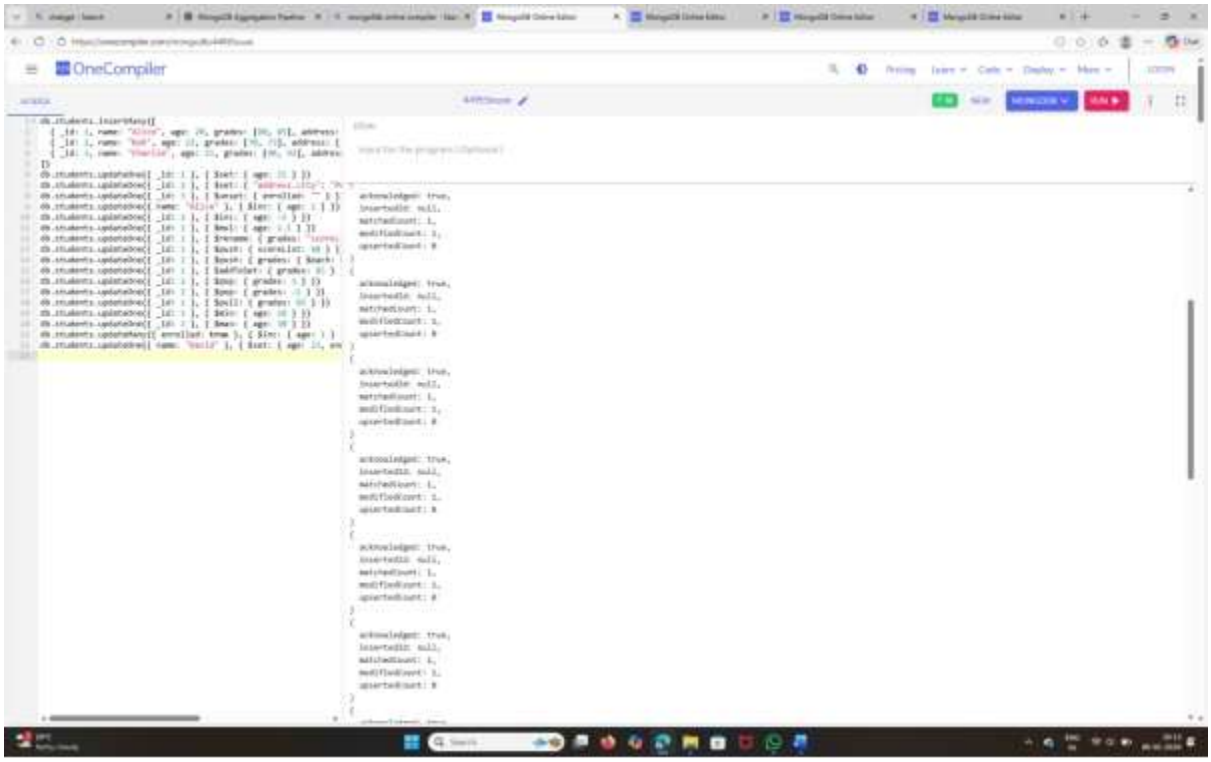
```
  { $pull: { grades: 85 } }
);
db.students.updateOne(
  { name: "Bob" },
  { $unset: { address: "" } }
);
db.students.updateOne(
  { name: "David" },
  {
    $set: {
      age: 20,
      grades: [60, 65],
      enrolled: true
    }
  },
  { upsert: true }
);
```

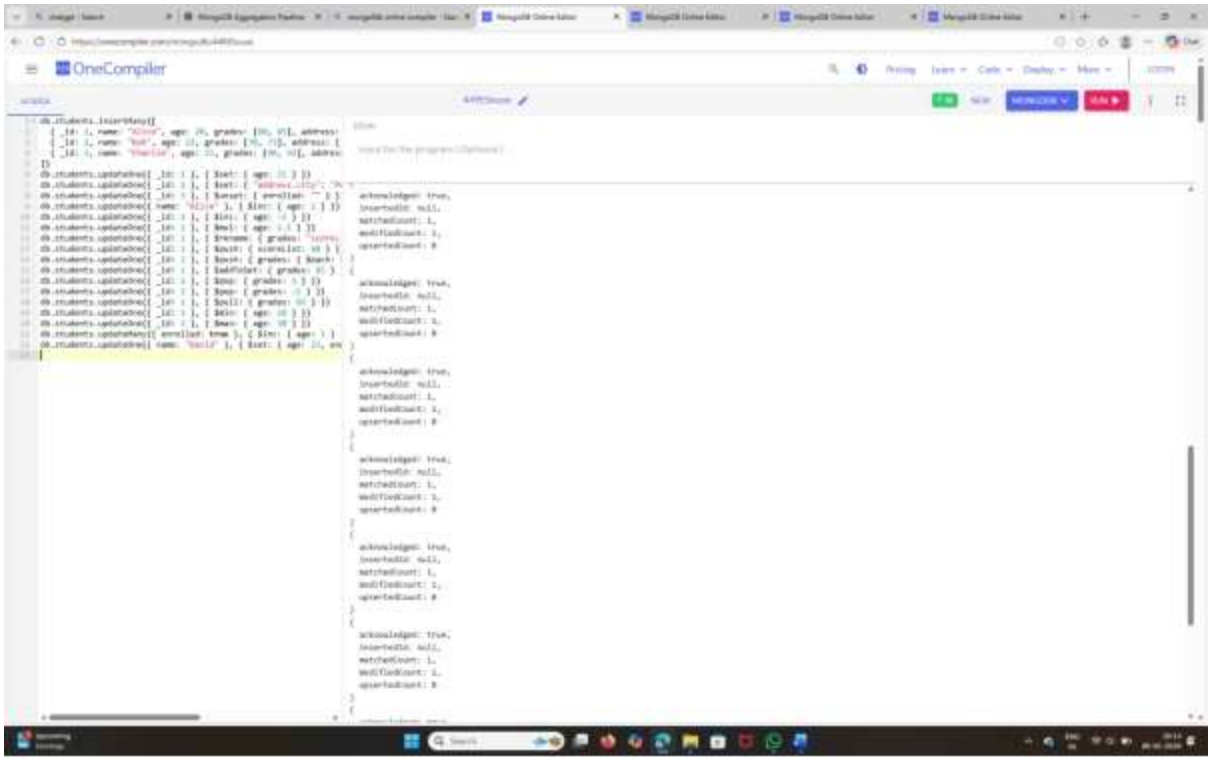
INPUT



OUTPUT



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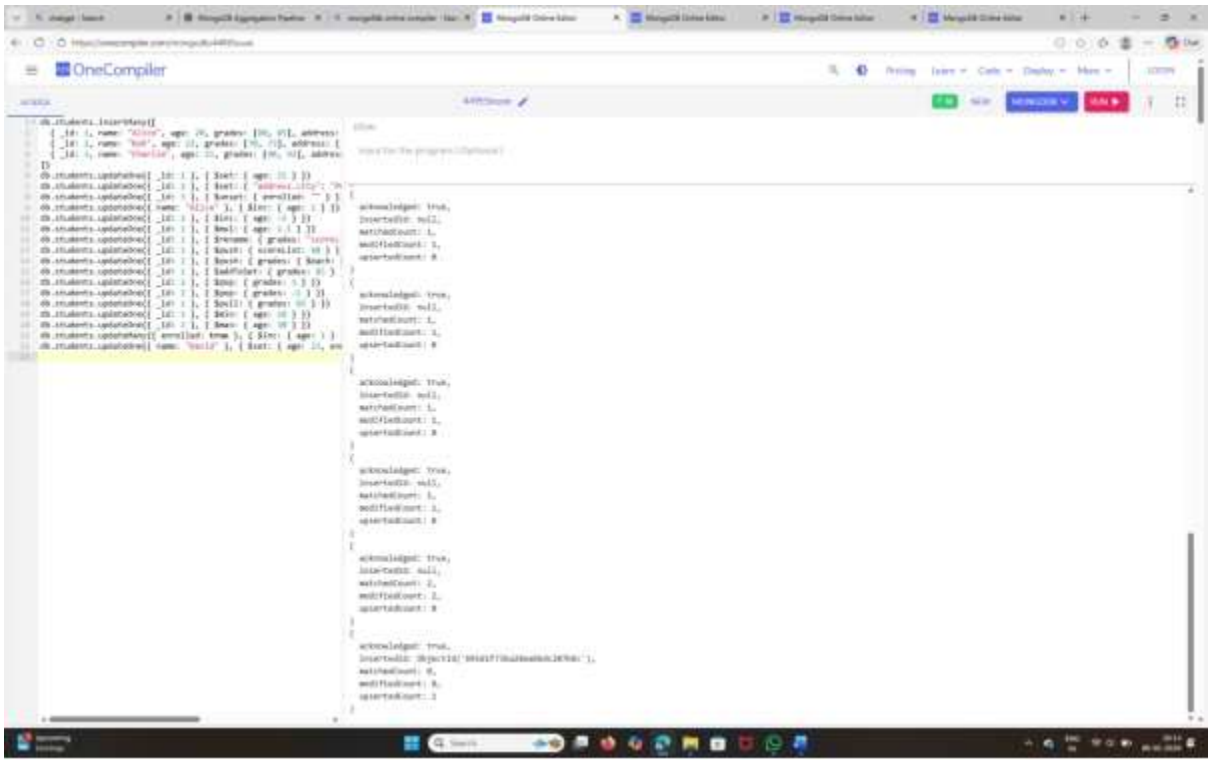
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EX.NO: 4

DATE: 09/12/2025

MongoDB Aggregation Pipeline (Match, Group, Project, Sort, Lookup, Out)

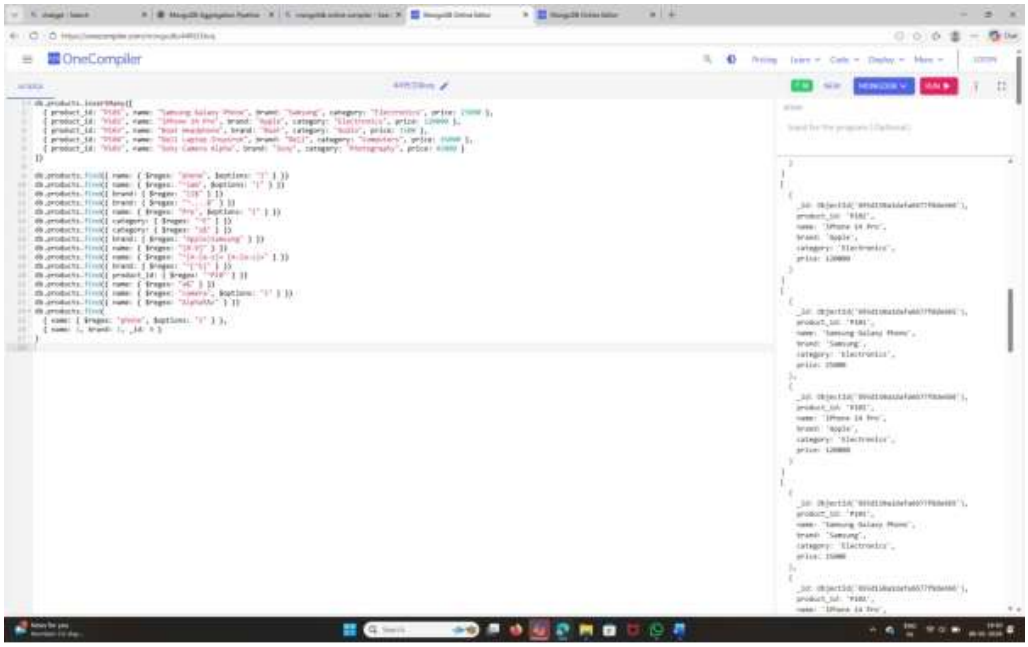
CODE:

```
db.students.insertMany([
  { studentId: 1, name: "Alex", age: 17, grade: 12, subject: "Math", marks: 85 },
  { studentId: 2, name: "Bella", age: 16, grade: 11, subject: "Math", marks: 78 },
  { studentId: 3, name: "Chris", age: 17, grade: 12, subject: "Science", marks: 92 },
  { studentId: 4, name: "Diana", age: 16, grade: 11, subject: "Science", marks: 88 },
  { studentId: 5, name: "Ethan", age: 17, grade: 12, subject: "Math", marks: 90 }
])

db.teachers.insertMany([
  { subject: "Math", teacher: "Mr. Smith" },
  { subject: "Science", teacher: "Ms. Johnson" }
])

db.students.aggregate([
  { $match: { marks: { $gt: 85 } } }
])

db.students.aggregate([
  {
    $group: {
      _id: "$subject",
      averageMarks: { $avg: "$marks" },
      totalStudents: { $sum: 1 }
```

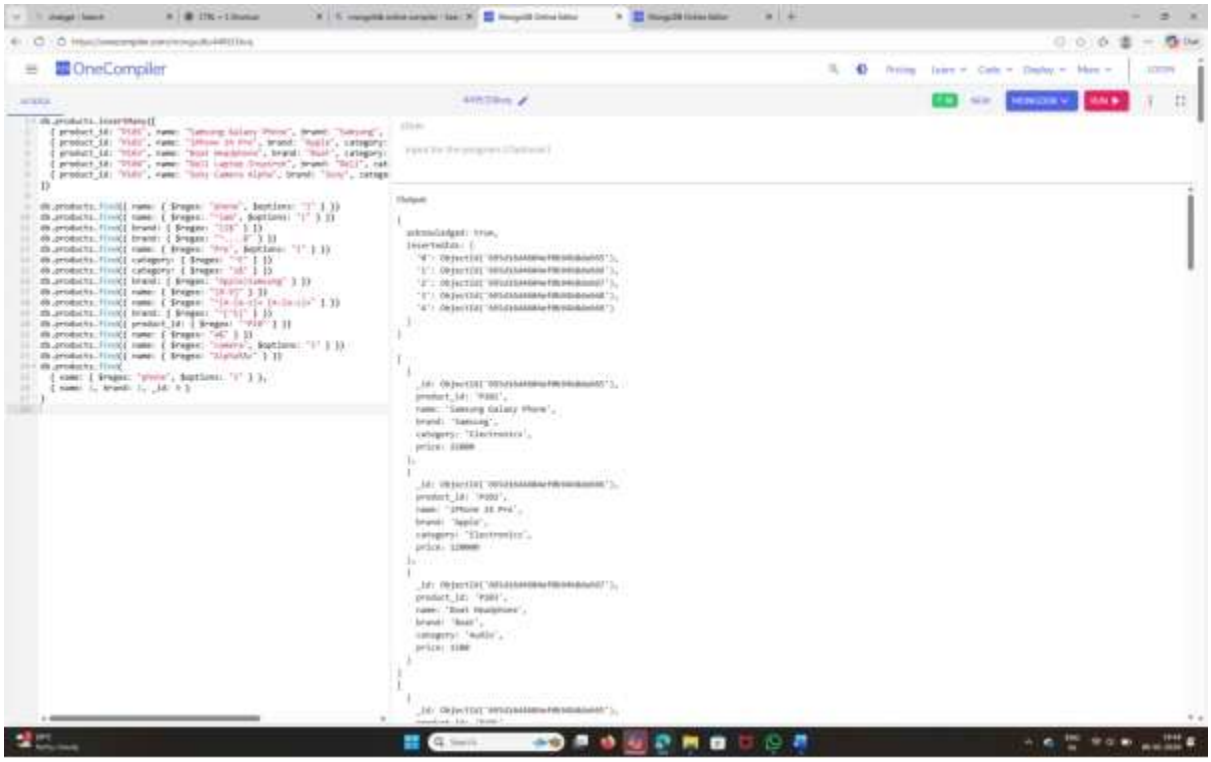
```
    }
  }
])

db.students.aggregate([
  {
    $project: {
      _id: 0,
      name: 1,
      subject: 1,
      marks: 1
    }
  }
])

db.students.aggregate([
  { $sort: { marks: -1 } }
])

db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 }
])

db.students.aggregate([
  {
    $addFields: {
      status: {
        $cond: {
          if: { $gte: ["$marks", 85] },
          then: "Pass",
```

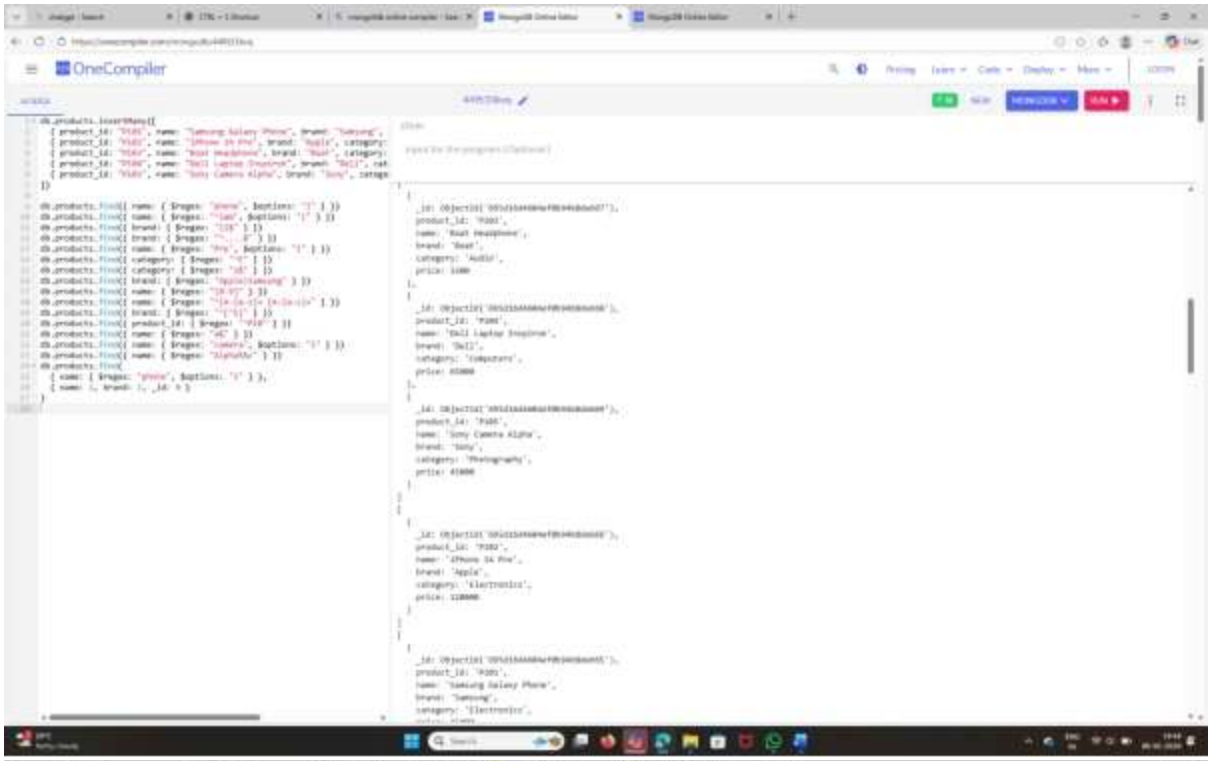
```
        else: "Fail"
      }
    }
  }
 }
])

db.students.aggregate([
 { $count: "totalStudents" }
])

db.students.aggregate([
 {
  $lookup: {
    from: "teachers",
    localField: "subject",
    foreignField: "subject",
    as: "teacherDetails"
  }
 }
])

db.students.aggregate([
 { $match: { marks: { $gte: 85 } } },
 { $out: "topStudents" }
])

db.students.aggregate([
 { $group: { _id: "$subject", avgMarks: { $avg: "$marks" } } },
 { $sort: { avgMarks: -1 } }
])
```

INPUT



OUTPUT

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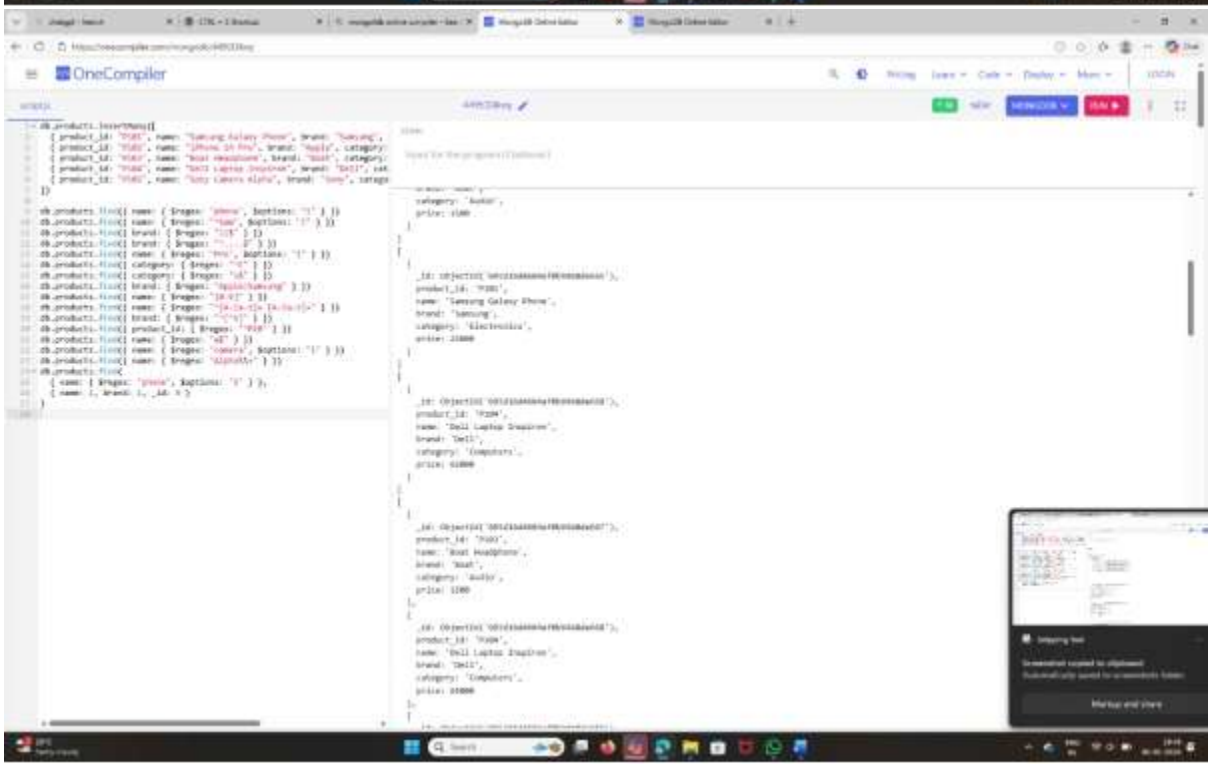
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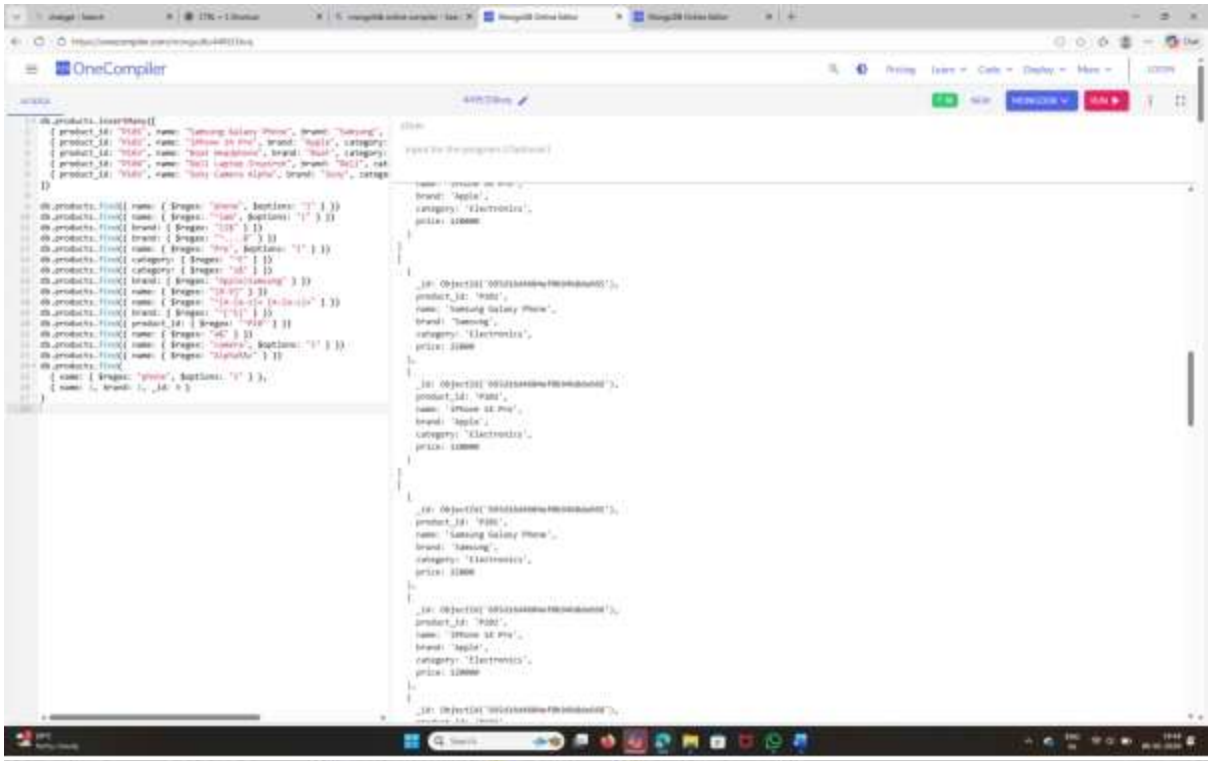
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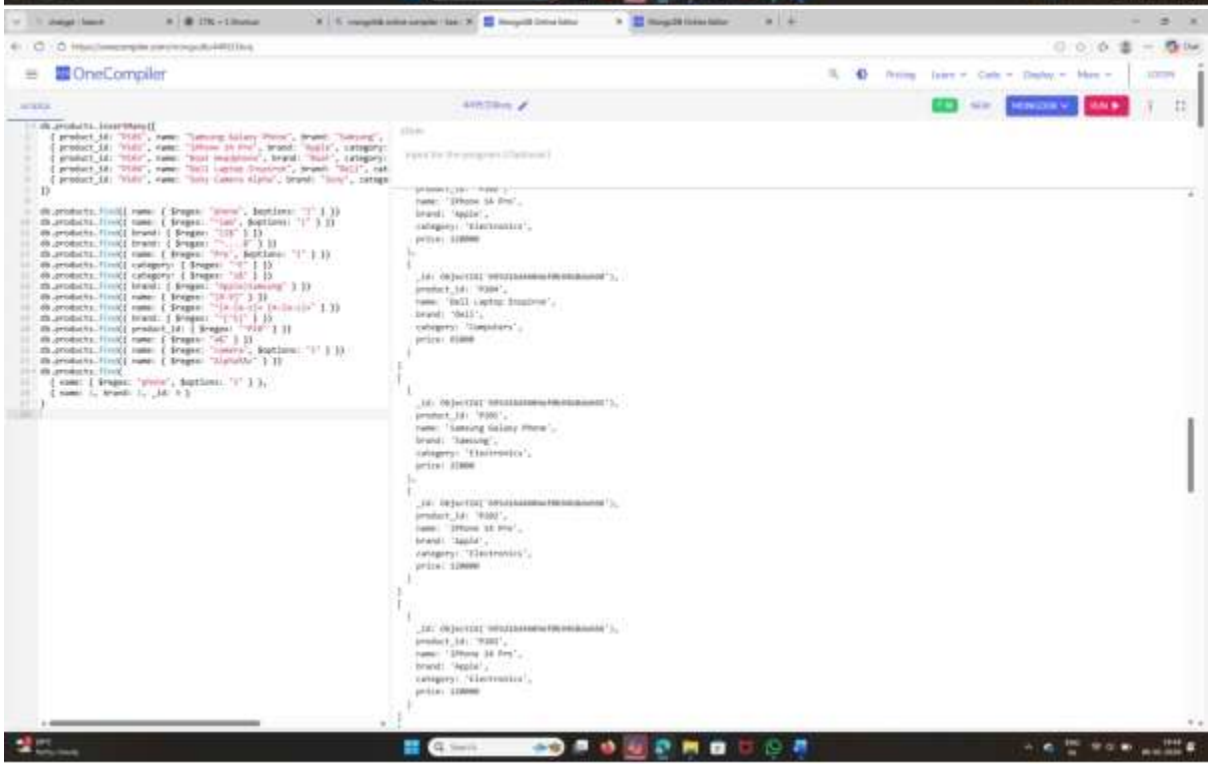
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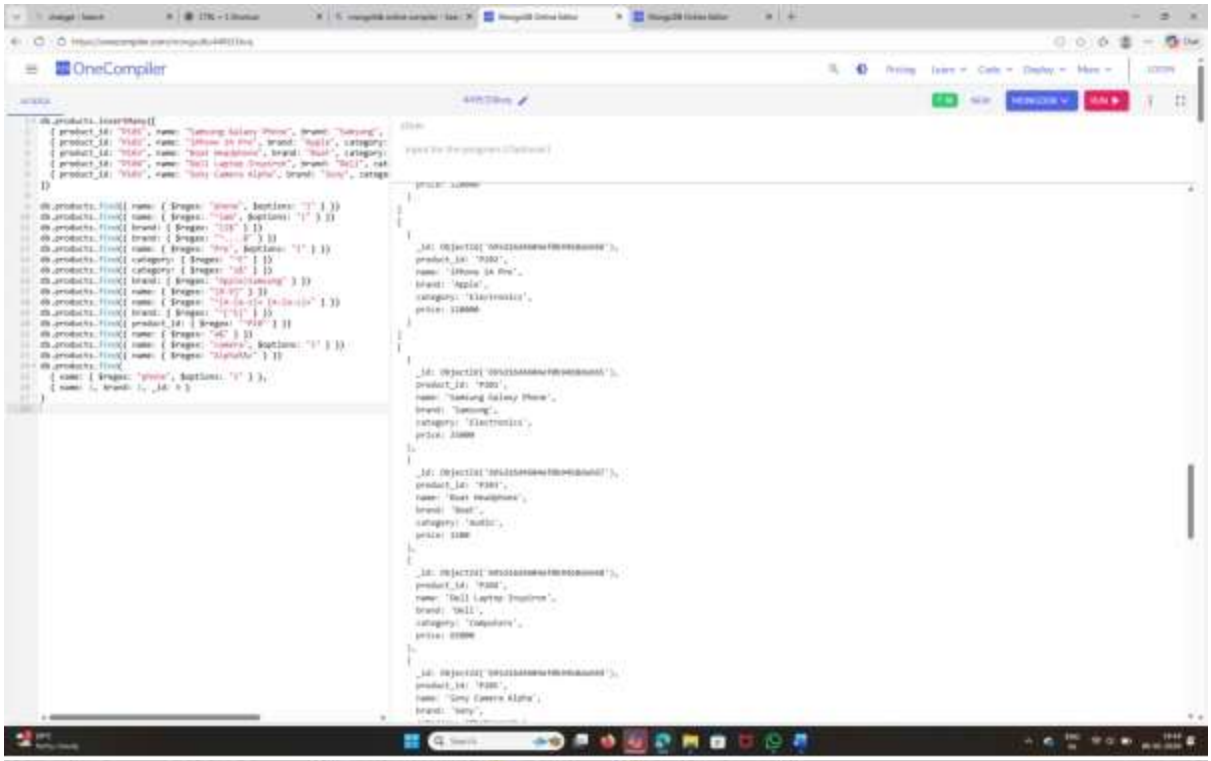
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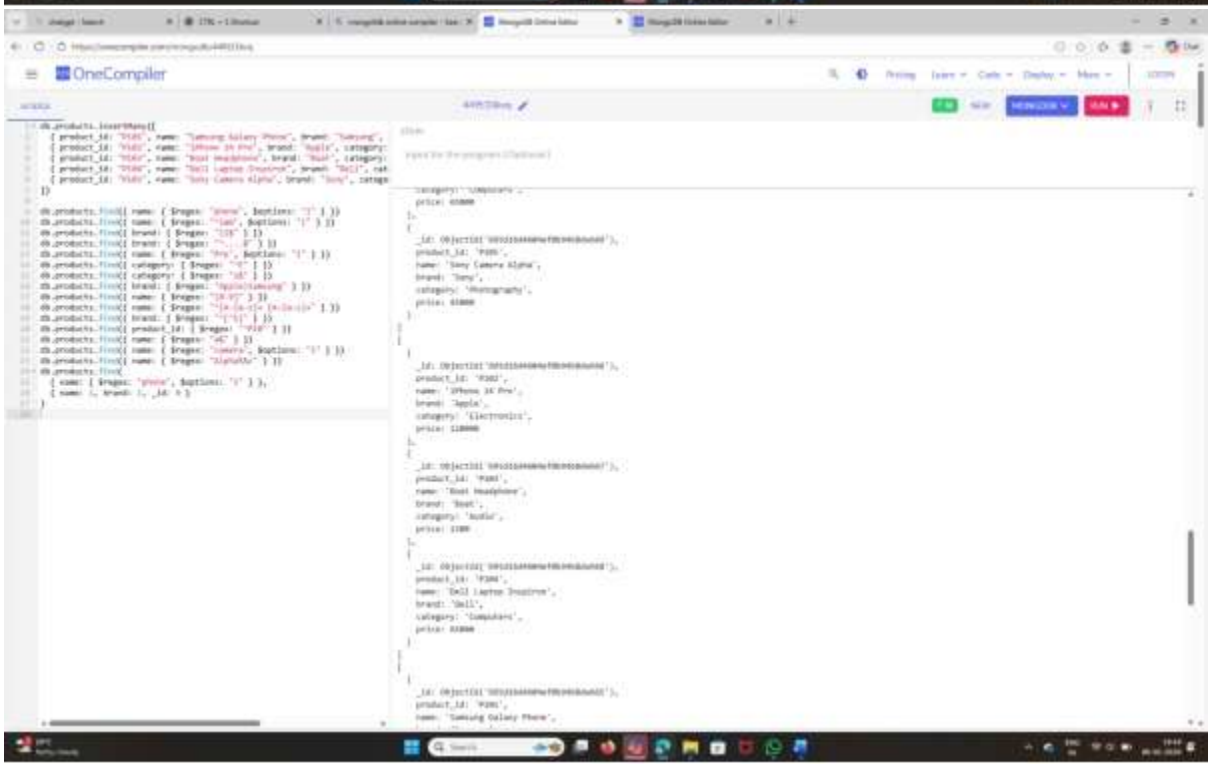
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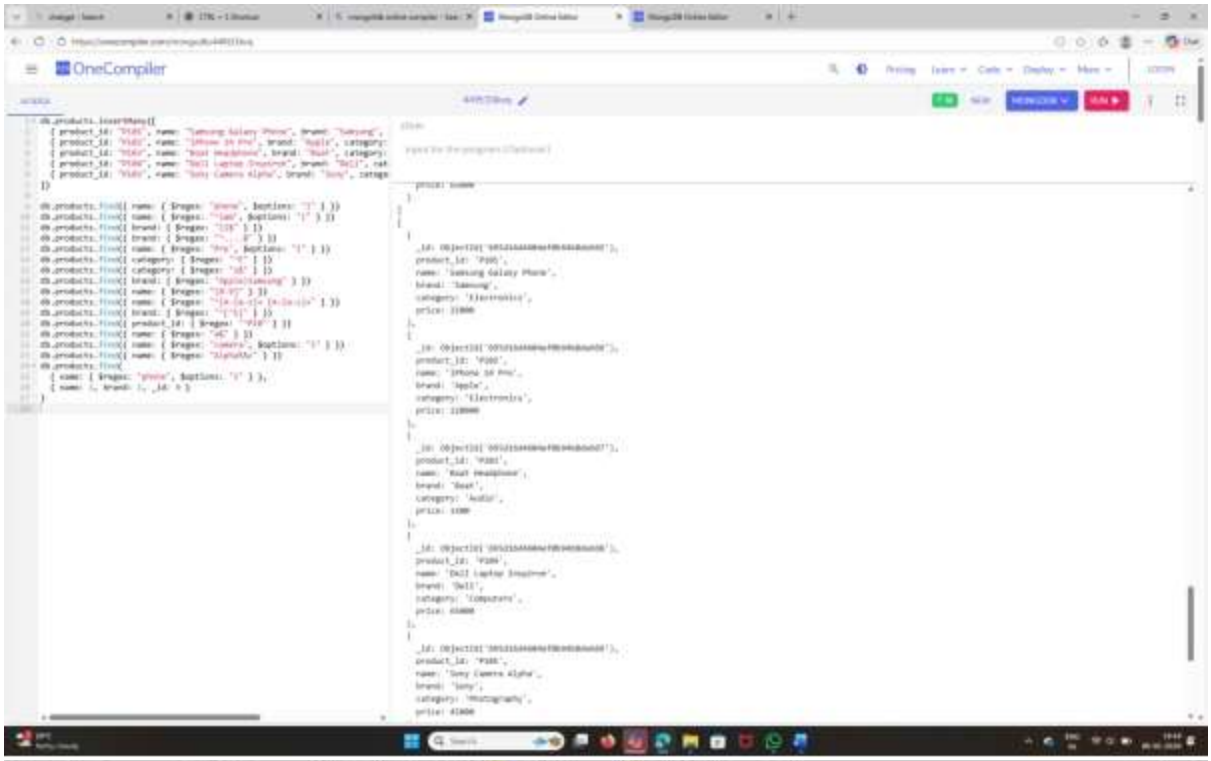
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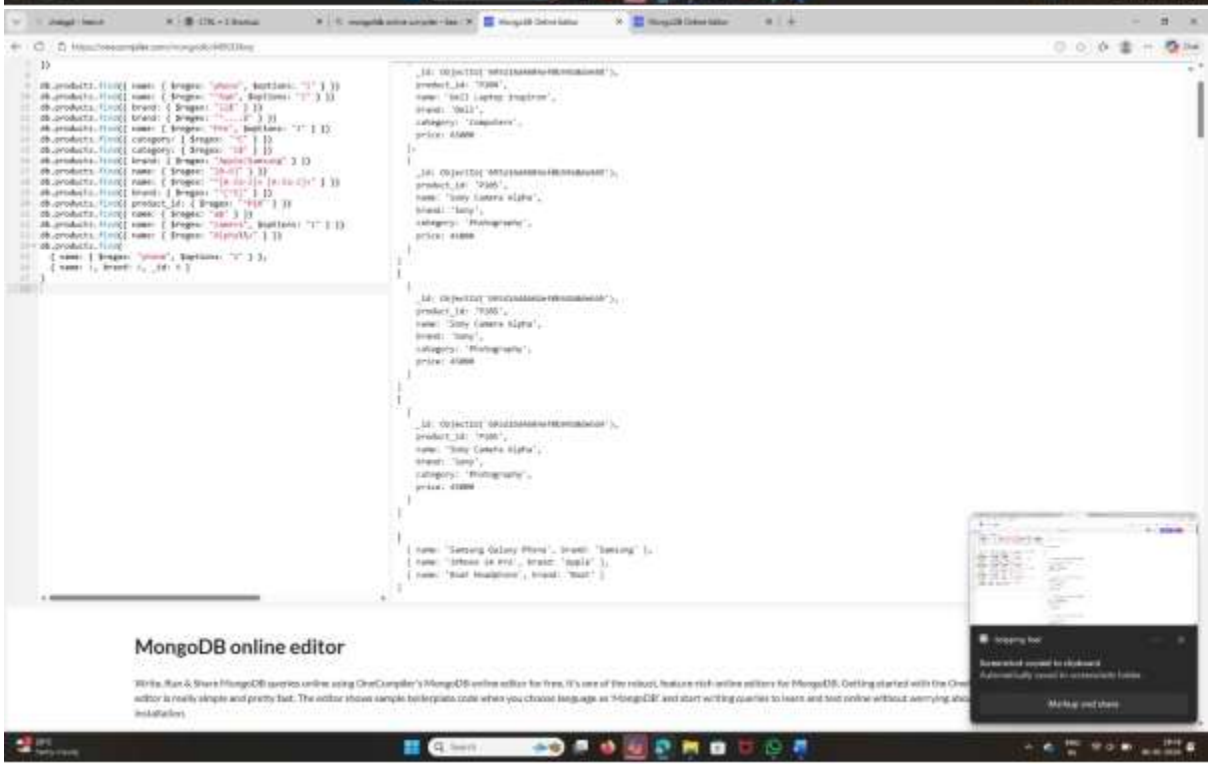

MongoDB online editor

EX.NO: 5

DATE: 23/12/2025

MongoDB Aggregation Pipeline Operations

(Using $match, $group, $project, $sort, $limit, $addFields, $count, $lookup, and $out)

CODE:

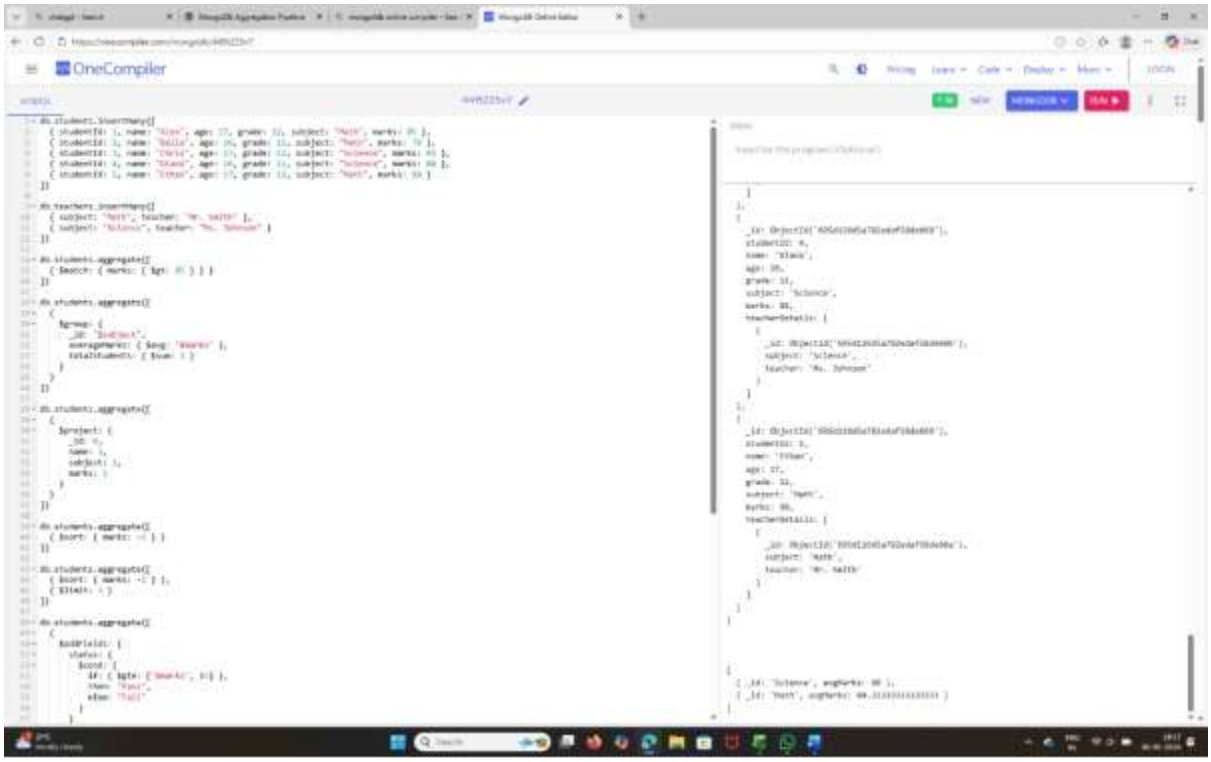
```
db.students.insertMany([
  { studentId: 1, name: "Alex", age: 17, grade: 12, subject: "Math", marks: 85 },
  { studentId: 2, name: "Bella", age: 16, grade: 11, subject: "Math", marks: 78 },
  { studentId: 3, name: "Chris", age: 17, grade: 12, subject: "Science", marks: 92 },
  { studentId: 4, name: "Diana", age: 16, grade: 11, subject: "Science", marks: 88 },
  { studentId: 5, name: "Ethan", age: 17, grade: 12, subject: "Math", marks: 90 }
])

db.teachers.insertMany([
  { subject: "Math", teacher: "Mr. Smith" },
  { subject: "Science", teacher: "Ms. Johnson" }
])

db.students.aggregate([
  { $match: { marks: { $gt: 85 } } }
])

db.students.aggregate([
  {
    $group: {
```

```
      _id: "$subject",
      averageMarks: { $avg: "$marks" },
      totalStudents: { $sum: 1 }
    }
  }
])

db.students.aggregate([
  {
    $project: {
      _id: 0,
      name: 1,
      subject: 1,
      marks: 1
    }
  }
])

db.students.aggregate([
  { $sort: { marks: -1 } }
])

db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 }
])

db.students.aggregate([
  {
    $addFields: {
      status: {
```

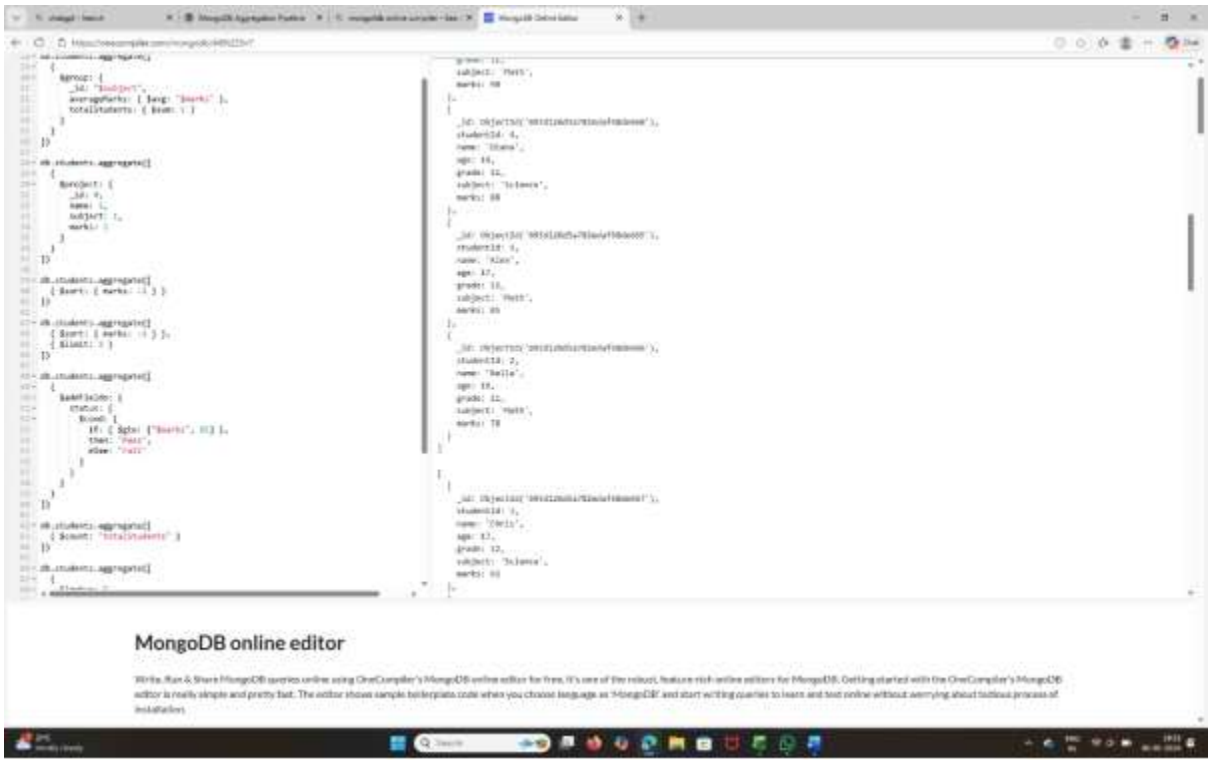
```
        $cond: {
          if: { $gte: ["$marks", 85] },
          then: "Pass",
          else: "Fail"
        }
      }
    }
  }
])

db.students.aggregate([
  { $count: "totalStudents" }
])

db.students.aggregate([
  {
    $lookup: {
      from: "teachers",
      localField: "subject",
      foreignField: "subject",
      as: "teacherDetails"
    }
  }
])

db.students.aggregate([
  { $match: { marks: { $gte: 85 } } },
  { $out: "topStudents" }
])

db.students.aggregate([
```

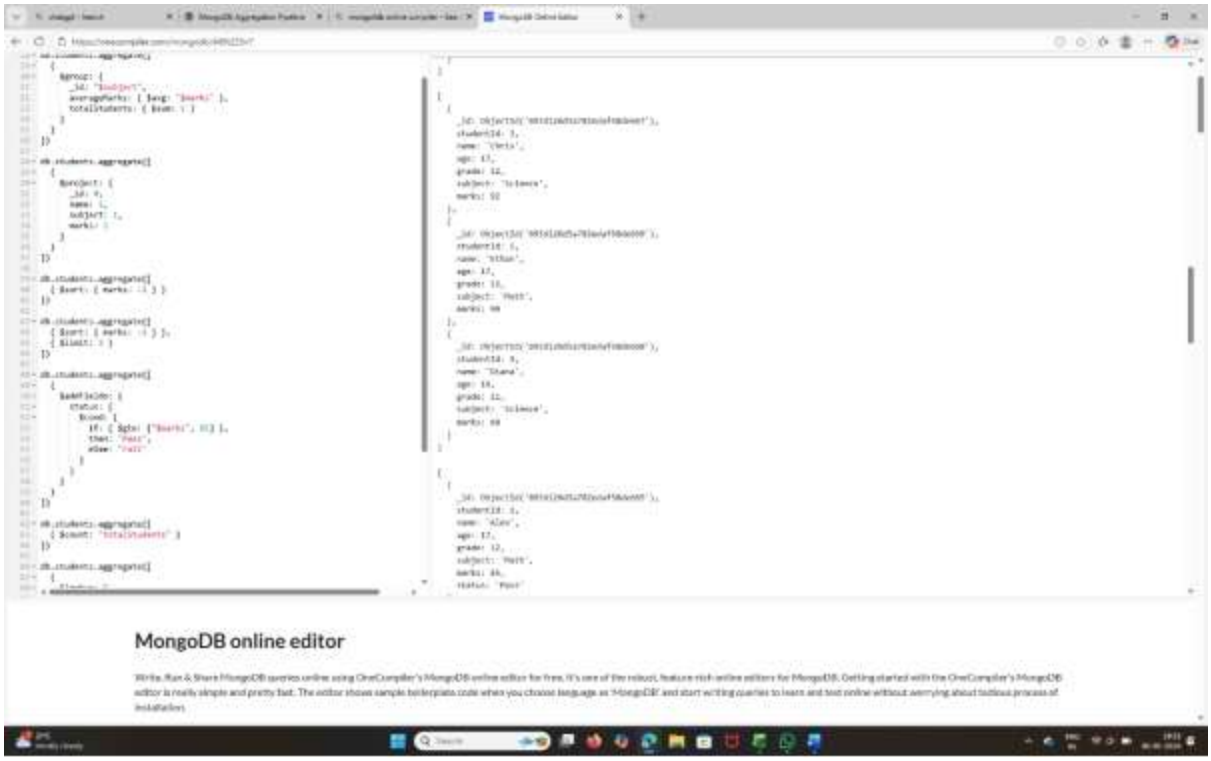
```
  { $group: { _id: "$subject", avgMarks: { $avg: "$marks" } } },
  { $sort: { avgMarks: -1 } }
])
```

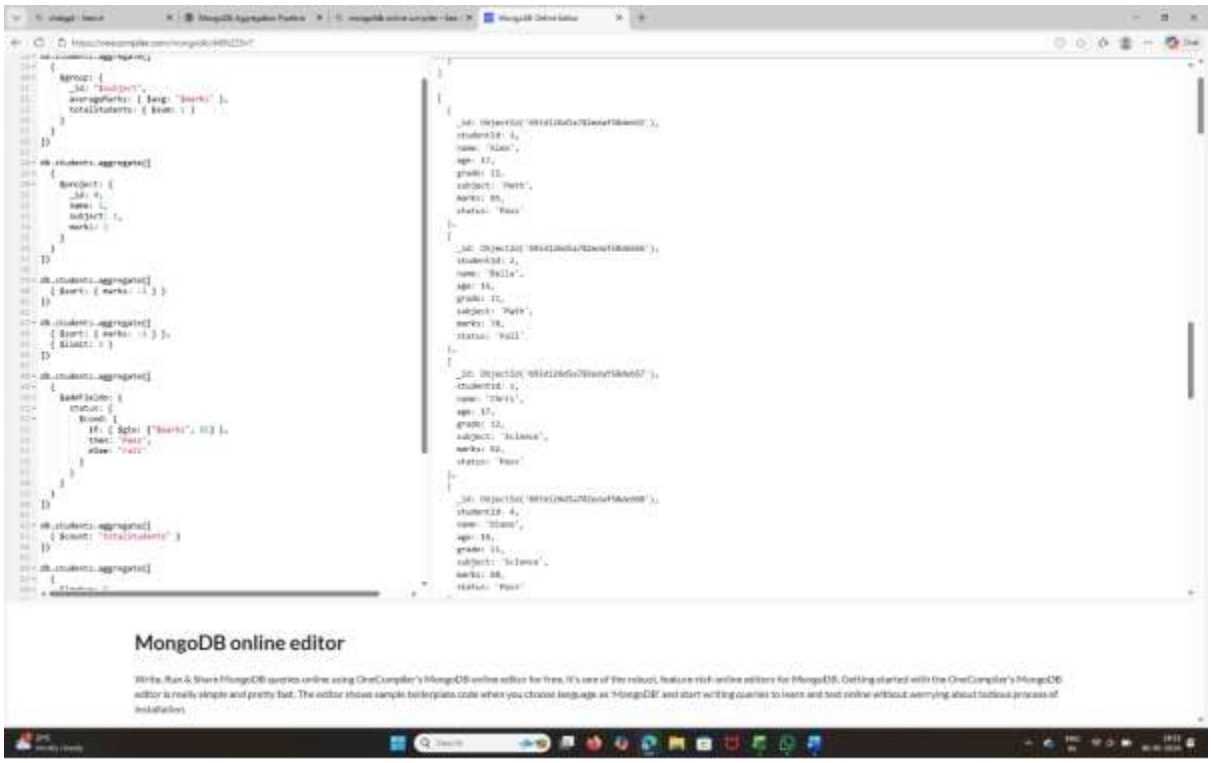
INPUT



OUTPUT

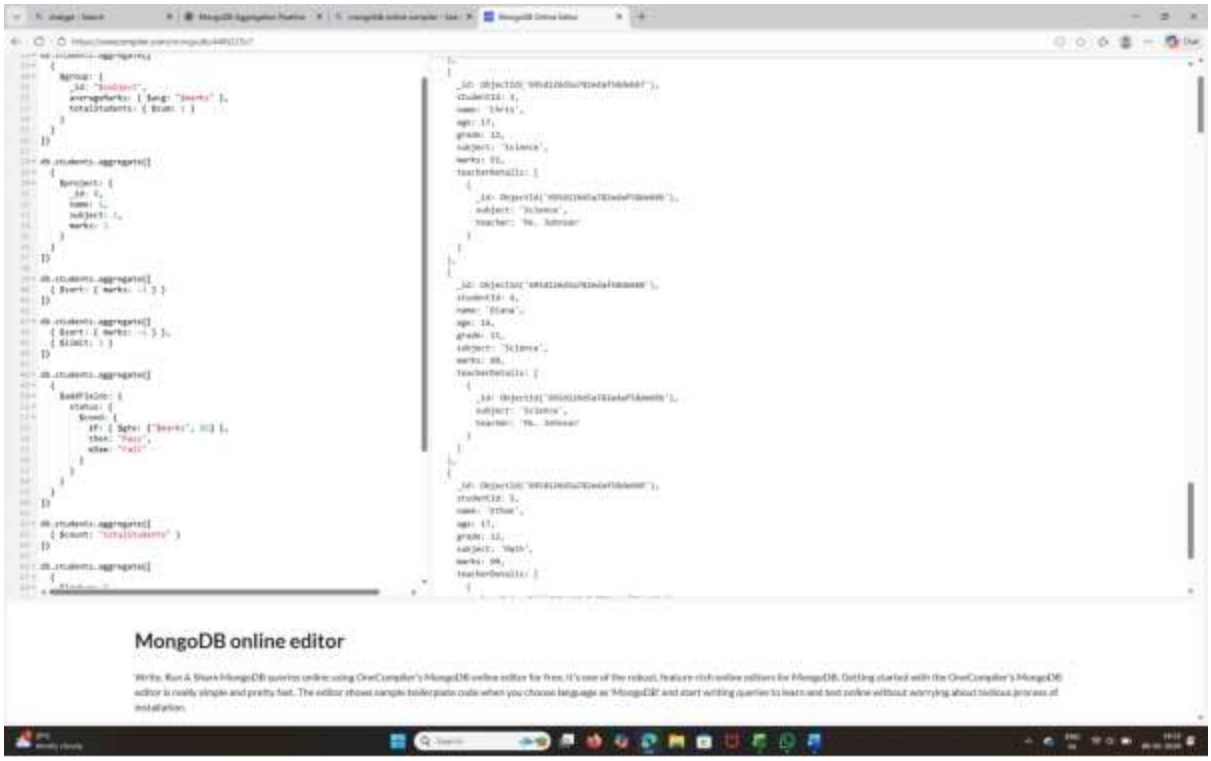



MongoDB online editor


MongoDB online editor

MongoDB online editor

MongoDB online editor


MongoDB online editor

MongoDB online editor

EX.NO: 6

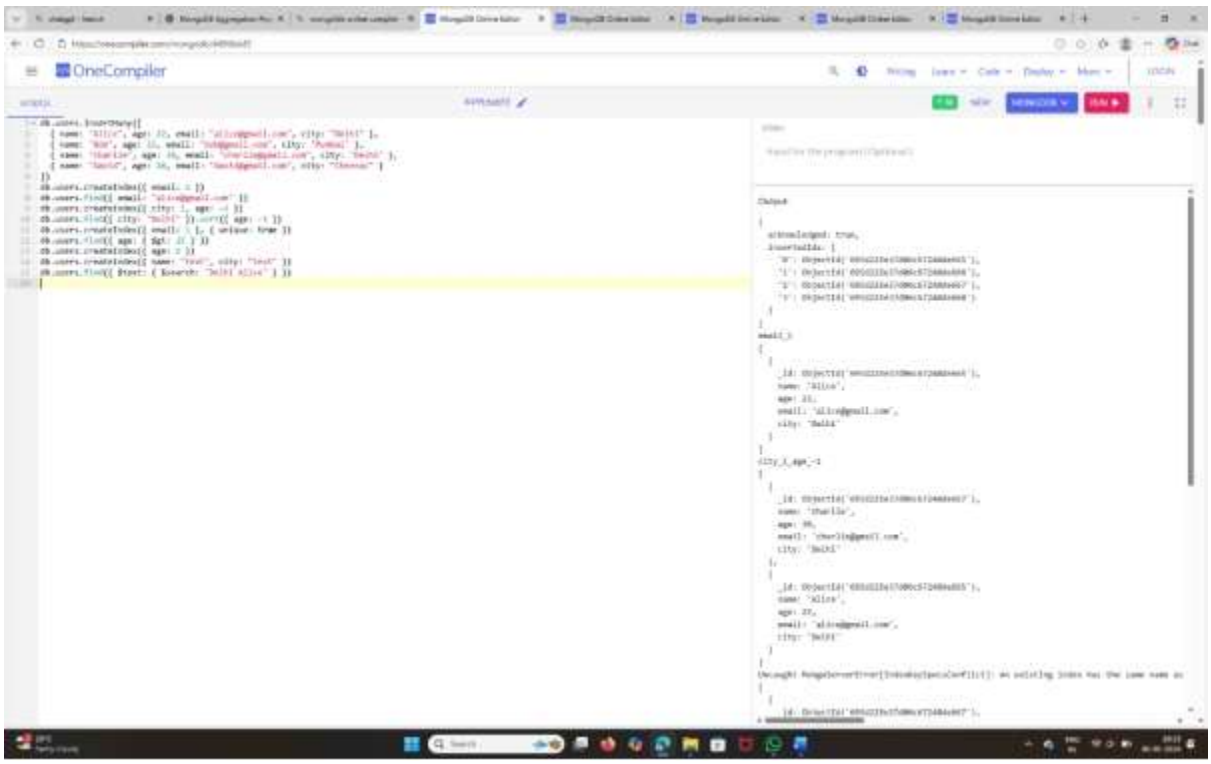
DATE: 30/12/2025

MongoDB Indexing (Single Field, Compound & Text Indexes)

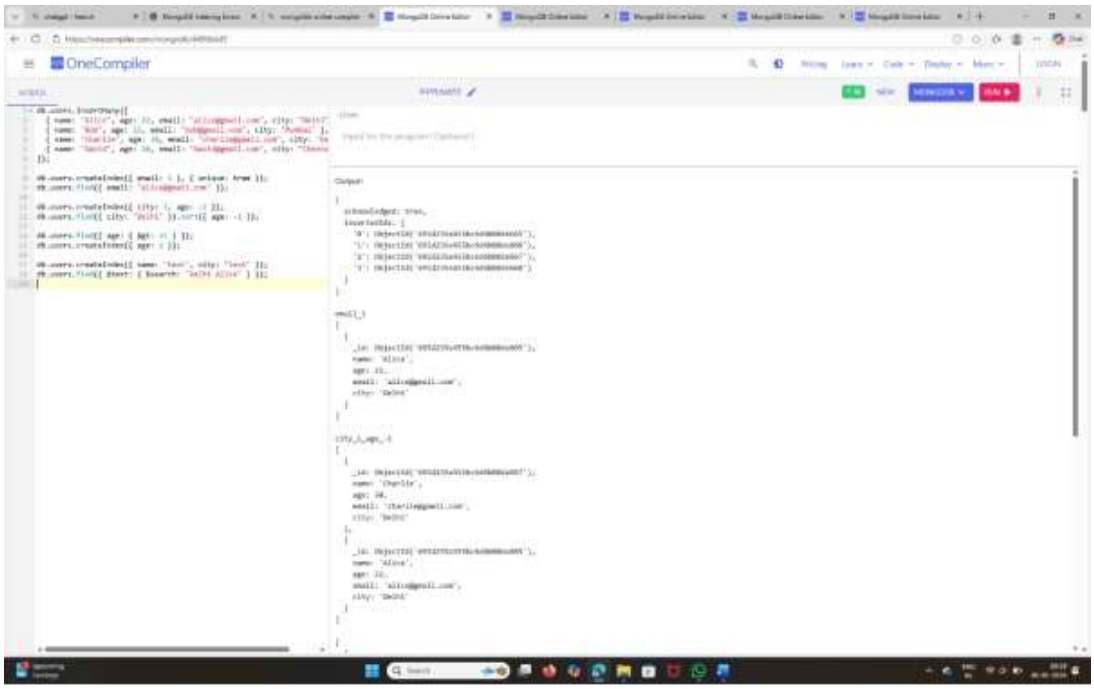
CODE:

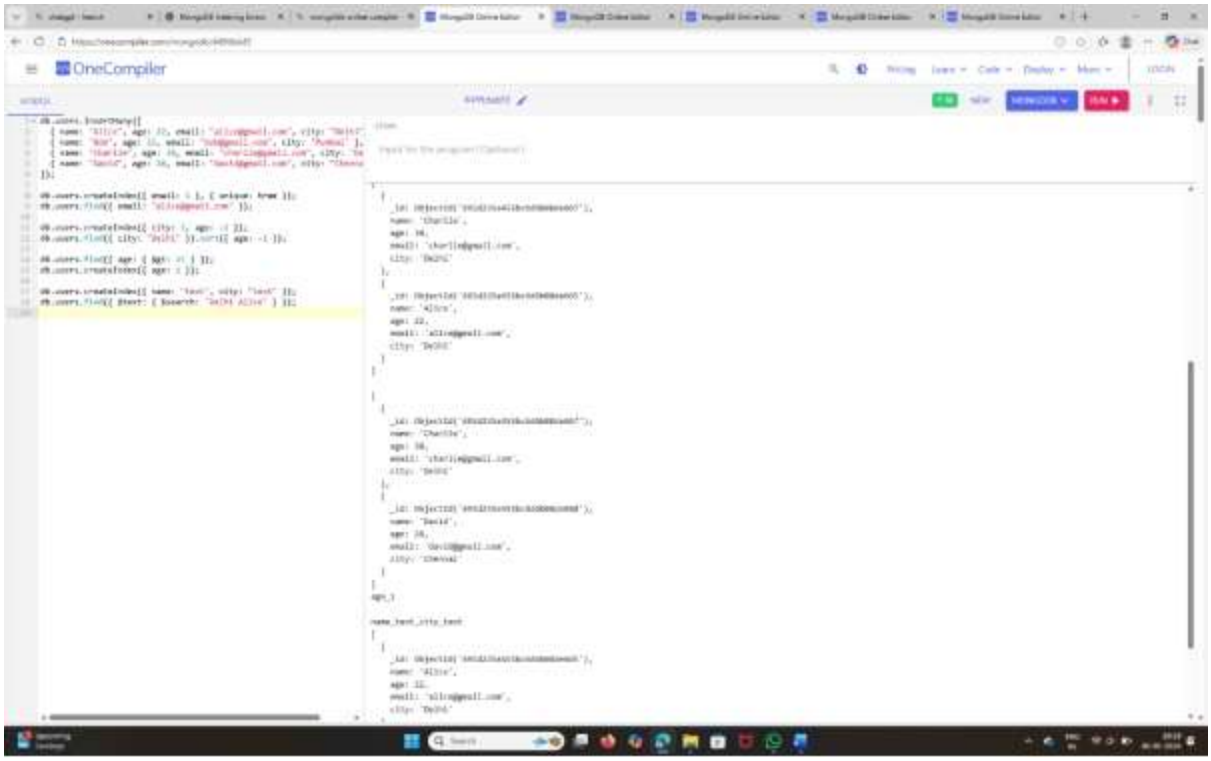
```
db.users.insertMany([
  { name: "Alice", age: 22, email: "alice@gmail.com", city: "Delhi" },
  { name: "Bob", age: 25, email: "bob@gmail.com", city: "Mumbai" },
  { name: "Charlie", age: 30, email: "charlie@gmail.com", city: "Delhi" },
  { name: "David", age: 28, email: "david@gmail.com", city: "Chennai" }
]);
db.users.createIndex({ email: 1 }, { unique: true });
db.users.find({ email: "alice@gmail.com" });
db.users.createIndex({ city: 1, age: -1 });
db.users.find({ city: "Delhi" }).sort({ age: -1 });
db.users.find({ age: { $gt: 25 } });
db.users.createIndex({ age: 1 });
db.users.createIndex({ name: "text", city: "text" });
db.users.find({ $text: { $search: "Delhi Alice" } });
```

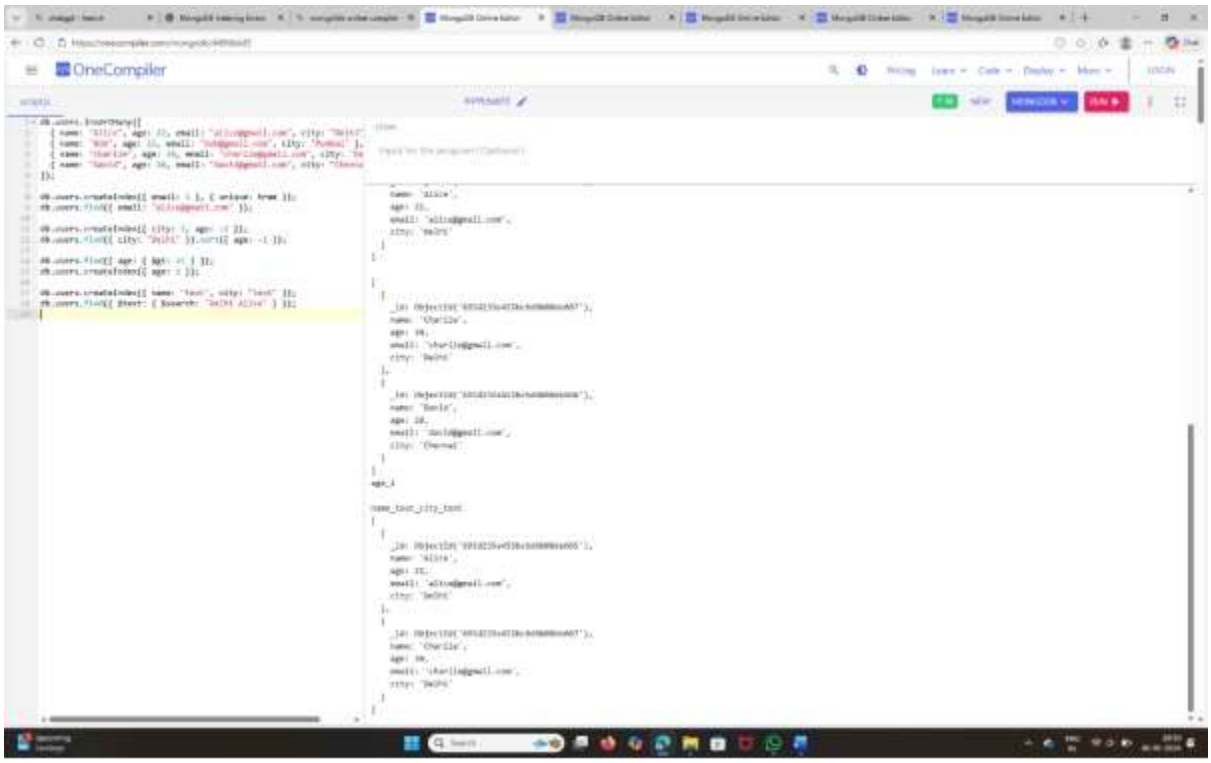
INPUT



OUTPUT




OneCompiler


OneCompiler

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